# The Decay Envelope of the Rvachev Sinc Product

## A comparative audit of four proposed solutions, with numerical adjudication and a corrected constant in the published literature

Comparative analysis for Vladimir Reshetnikov's `ProveIt` project
covering the four drafts in `drafts/rvachev_up_fourier_decay/`

26 August 2026

**Abstract**

Mathematics Stack Exchange question 2745497 asks for a positive, analytic, monotone decreasing gauge $g$ such that the Cesàro mean of $|f|/g$ tends to one, where $f(x) = \prod_{n\geq 0} \operatorname{sinc}(\pi x/2^n)$ is the Fourier transform of Rvachev's up-function. Four AI-produced documents in this directory propose solutions. This audit verifies their overlapping claims numerically and against the literature, adjudicates their one genuine mutual contradiction, and states what the answer to the original question actually is.

The four documents agree, correctly, on the universal factor $\exp(-(\log x)^2/(2\log 2))$ and on a spectrum of power corrections $x^{-\kappa}$ that depends on the meaning of "amplitude": $\kappa_\infty = 2.3590148791\ldots$ for the extremal envelope, $\kappa_2 = \log_2(2\pi) = 2.6514961295\ldots$ for the quadratic mean, $\kappa_1 = 2.7480700149\ldots$ for the arithmetic (Cesàro) mean, and $\kappa_0 = 3.1514961295\ldots$ for Lebesgue-typical points. Documents 3 and 4 answer the literal question: the gauge $g(x) = A_1 x^{-\kappa_1} \exp(-(\log x)^2/(2\log 2))$ with $\kappa_1 = \frac{1}{2} + \log_2(\pi/\varrho_1) = 2.748070014871334\ldots$, where $\varrho_1 = 0.661322602060565\ldots$ is the Perron root of an explicit transfer operator, satisfies the two-sided bounded form of the question and attains the Cesàro limit 1 along $t = 2^k$; but we prove that no gauge whose within-shell shape stabilizes—in particular no Hardy-field ("elementary") gauge—can attain the limit along all $t$: the limit points trace a nonconstant dyadic-mantissa profile (computed here to span $[0.9249, 1.1133]$), and constancy of the profile would force the transfer-operator eigenmeasure to be absolutely continuous, contradicting its singularity, which we prove from the elementary bound $\varrho_1 \geq 3^{-1/2} > \frac{1}{2}$. We add one repair, with proof: with the same gauge, the *logarithmic* mean $\frac{1}{\log(t/t_0)} \int_{t_0}^{t} \frac{|f(x)|}{g(x)} \frac{\mathrm{d}x}{x}$ converges for all $t \to \infty$, so the multiplicative form of the question has an exact affirmative answer.

The audit's principal forensic finding concerns the law of the iterated logarithm. Document 1 derives, by a martingale argument, per-step variance $\pi^2/4$ and hence the fluctuation constant $\pi/\sqrt{2}$; Document 3 quotes the published Theorem 1.2 of Aistleitner–Hofer–Larcher (Ann. Inst. Fourier 67 (2017) 637–687), whose constant is $\pi$. Both cannot be right. An exact Parseval computation of the paper's own variance functional (2.9)—confirmed by measurement over exact rational orbits, $\operatorname{Var}(S_{1000}) = 2459.5$ against the predictions 2464.1 for variance $\pi^2/4$ and 4928.2 for variance $\pi^2/2$—shows that Document 1 is correct and that the published equation (4.5) omits the Parseval factor $\frac{1}{2}$: the constants in the published Theorems 1.1 and 1.2 should each be divided by $\sqrt{2}$.

Two interpretations are developed beyond what any of the four documents attempts. The envelope-of-lobe-maxima reading (section 6): the peak sequence $E_n$ reproduces the decay spectrum in miniature—limsup envelope $\kappa_\infty$, zero liminf on that scale, Cesàro exponent $\kappa_1$ via a new peak–integral comparison lemma (with the measured rigidity $\sum_{\text{shell}} E_n / \int_{\text{shell}} |f| = 1.9237$, stable to five digits across six octaves), and a density-one $\kappa_0$ law with a central limit theorem whose finite-size right-tail truncation is explained quantitatively by the Gelfond cap. The root-mean-square reading (section 7): the $q = 2$ layer is exactly solvable—the transfer operator has the closed subleading eigenvalue $-\frac{1}{4}$ with eigenfunctions $\sin 2\pi x$ and $1 + 3\cos 2\pi x$, making the RMS envelope constant $A_2 = 0.103008918$ converge at the alternating geometric rate $(-\frac{1}{2})^k$—yet the all-$t$ Cesàro limit still fails (singular eigenmeasure, profile $[0.894, 1.192]$), and the total energy check $\int_{\mathbb{R}} f^2 = \frac{1}{2} + \sum_m f(m + \frac{1}{2})^2 = 0.808873535967972$ closes to $2 \cdot 10^{-15}$.

# Contents

# 1 Purpose, inputs, and method

This document audits the four solution drafts

| label | file |
|---|---|
| Document 1 | `rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex` |
| Document 2 | `Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex` |
| Document 3 | `Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex` |
| Document 4 | `rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex` |

which respond to the problem posed in the adjacent OCR copy of Mathematics Stack Exchange question 2745497 [6] and, indirectly, to the now-deleted repository draft *Decay rate conjecture* (git object `e1cdd6260`, quoted in section 11.1).

The audit method follows the repository's engineering policy, with one governing principle: **every adjudication in this document rests on a proof; numerical experiments serve only as insight and as validation of conclusions reached rigorously.** Where a verdict is stated (theorems 5.5, 5.7, 6.2, 7.2, 9.1 and 9.2, finding 9.3), a proof or a complete proof architecture is given, with the imported classical inputs named explicitly (section 5.2). Independently of that, every quantitative sentence was *computed*, not reasoned to: all numerical values were produced by the Python scripts reproduced in section B, run on 26 August 2026, and independent computations (direct quadrature versus spectral prediction, direct maximization versus published tables, Monte Carlo versus closed-form variances) are compared digit by digit in section 10. Literature quotations from Aistleitner–Hofer–Larcher [1] were checked against three independent sources: the publisher's PDF *as rendered page images* (pages 641, 651, 663), the OCR transcription stored at `docs/papers/AIF_2017__67_2_637_0/`, and the authors' own LaTeX source of the arXiv version at `docs/papers/arXiv-1502.06738v2/Lacunary_products.tex`. All three agree on every formula quoted here, so no OCR artifact affects the findings.

# 2 The object and the question

Throughout, $\operatorname{sinc} z = \sin z/z$, $\operatorname{sinc} 0 = 1$, $a = \log 2$, and

$$f(x) = \prod_{n=0}^{\infty} \operatorname{sinc}\Big(\frac{\pi x}{2^n}\Big), \qquad x \in \mathbb{R}. \tag{1}$$

Under the convention $\widehat{h}(\xi) = \int h(t)\mathrm{e}^{-2\pi\mathrm{i}\xi t}\,\mathrm{d}t$ one has $f = \widehat{\mathrm{up}}$, the Fourier transform of Rvachev's up-function; in the Lean development this object is `rvachevFourier_eq_product` with refinement law `rvachevFourier_scaling`. The function is entire, even, vanishes at each nonzero integer $m$ with multiplicity $1 + v_2(|m|)$, has exactly one critical point between consecutive integer zeros, and its sign on $(m, m+1)$ is the signed Thue–Morse value $(-1)^{s_2(m)}$ (all four documents prove these; the sign statement is also in the Thue–Morse Formula Atlas, §“The sign of the Fabius sinc product”).

The question [6] asks for a real $g$, “elementary, if possible … analytic, positive and monotone decreasing (and having monotone derivatives of any order, if possible)”, such that

$$\lim_{t\to\infty} \frac{1}{t-t_0}\int_{t_0}^{t} \frac{|f(x)|}{g(x)}\,\mathrm{d}x = 1 \tag{Q2}$$

for large enough $t_0$, or at least

$$\exists\, 0 < A < B\ \forall t > t_0 : \quad A < \frac{1}{t-t_0}\int_{t_0}^{t} \frac{|f(x)|}{g(x)}\,\mathrm{d}x < B. \tag{Q3}$$

The asker’s empirical guess was “something close to $\exp(-\log^2 x)$, but perhaps not exactly that”.

Two remarks on reading the question. First, the Cesàro mean in (Q2)–(Q3) makes this an $L^1$ (arithmetic-mean) question: $g$ is asked to be the *mean local amplitude* of $|f|$, not a pointwise envelope, not an envelope of the lobe maxima, and not a root-mean-square amplitude. Much of what the four documents prove concerns those other readings; section 3 maps out how the answer changes with the reading. Second, since $\int_{t/2}^{t}$ carries half the weight of $\int_{t_0}^{t}$, condition (Q2) is equivalent to the statement that the *partial-shell* averages $\frac{1}{2^k}\int_{2^k}^{2^k y} |f|/g \, \mathrm{d}x \to y-1$ for every $y \in [1,2]$; this equivalence is what ultimately decides (Q2) negatively for natural gauges (section 5).

## 3 What “asymptotic decay rate” can mean

All four documents, correctly, refuse to assign the product a single decay rate. After the universal factor

$$\mathcal{E}(x) = \exp\left(-\frac{(\log x)^2}{2\log 2}\right),$$

whose exponent’s coefficient $-1/(2\log 2) = -0.7213475\ldots$ comes from the deterministic denominator $\prod_{j\le \log_2 x} 2^j$, the surviving power of $x$ depends on the interpretation. The verified dictionary:

| Interpretation | power $\kappa$ after $\mathcal{E}(x)$ | status |
|---|---|---|
| pointwise equivalent $\lvert f\rvert \sim g$ | none | impossible: integer zeros; and on a.e. dyadic ray the centered logarithm has unbounded iterated-logarithm excursions |
| envelope of *all* lobe maxima $E_n$ | none | impossible even after passing to local maxima: the peaks straight after $2^k$ decay with the doubled quadratic coefficient $-1/\log 2$ (Docs. 2, 4; verified section 10.5); full four-layer analysis in section 6 |
| largest lobe in $[2^k, 2^{k+1}]$ | $\kappa_\infty =$ 2.3590148791 | sharp as a limsup; the true annular maximum carries a second log-normal factor $\exp(-(\log\log x)^2/(2\log 2) + \cdots)$ (Docs. 1, 2; verified section 10.4) |
| Lebesgue-typical dyadic ray | $\kappa_0 =$ 3.1514961295 | holds a.e. with $o(\log x)$ error; the error is genuinely random, of iterated-logarithm size (Docs. 1–4; constant of section 9) |
| arithmetic mean = the literal question | $\kappa_1 =$ 2.7480700149 | (Q3) solvable; (Q2) solvable along $t = 2^k$ and for the logarithmic mean, not for all $t$ (Docs. 3, 4; section 5) |
| root mean square | $\kappa_2 = \log_2(2\pi) =$ 2.6514961295 | exact identity $\int_0^1 \prod_{j<n} \sin^2(\pi 2^j t)\,\mathrm{d}t = 2^{-n}$; this is precisely the exponent of the deleted draft (Docs. 3, 4); exactly solvable layer, developed in section 7 |

Document 4 embeds the middle rows in a full $L^q$ spectrum $\kappa_q = (\log \pi + \frac{a}{2} - \log \Lambda_q)/a$ with $\Lambda_q = \varrho_q^{1/q}$ the $q$-th root of the Perron root of the weighted doubling transfer operator; $\kappa_q$ decreases strictly from $\kappa_0$ to $\kappa_\infty$. The verified values are collected in section A.

Two further interpretation issues are settled by the problem statement itself. The convention $\operatorname{sinc} z = \sin z / z$ (argument already containing $\pi$) is forced by the stated zero set $\mathbb{Z}_{>0}$ and by the plot in the question. And the normalization matters only through trivial rescalings: $f(x) = \widehat{\mathrm{up}}(x)$ exactly in the $\mathrm{e}^{-2\pi i x t}$ convention, so every statement transfers verbatim to the up-function literature.

Finally, the two forms (Q2) and (Q3) are genuinely different problems: (Q3) is solved by an explicit elementary gauge, while (Q2) fails for every gauge in the natural class (theorem 5.1). The choice of *averaging operator* is thus itself an interpretation with mathematical content: arithmetic averaging does not respect the multiplicative self-similarity $f(2x) = \operatorname{sinc}(2\pi x) f(x)$, and the obstruction to (Q2) disappears the moment the average is taken multiplicatively.

## 4 The common core, verified

All four documents rest on the same exact identity, stated here in the normalization of Document 3. Write $x = 2^k y$, $1 \le y < 2$, $t = y - 1$, $n = k + 1$,

$$P_n(t) = \prod_{j=0}^{n-1} \left|\sin(\pi 2^j t)\right|, \qquad H(y) = \prod_{m=1}^{\infty} \left|\operatorname{sinc} \frac{\pi y}{2^m}\right|.$$

Then

$$|f(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-(k+1)} H(y)\, P_{k+1}(y-1). \tag{2}$$

Numerical check: for $k \in \{3, 7, 12\}$ and random $y$, the two sides of (2) agree to $1.6 \cdot 10^{-12}$ in the logarithm (dominated by quadrature-free floating-point accumulation). The identity reduces every averaged decay question to the lacunary product $P_n$ under the doubling map $Tt = 2t \bmod 1$, and the elementary algebra

$$-\tfrac{a}{2} k(k+1) - (k+1)(c + \log y) = -\frac{L^2}{2a} - \left(\tfrac{1}{2} + \tfrac{c}{a}\right) L + \Phi_c(\log y), \qquad L = \log x,$$

(Document 3's identity, reproved symbolically) converts each per-level numerator scale $\mathrm{e}^{-c}$ into the power $\kappa = \frac{1}{2} + c/a$. The problem statement's own finite expansion

$$\prod_{n=0}^{m} \operatorname{sinc} \frac{\pi x}{2^n} = \frac{2^{\binom{m}{2}}}{(\pi x)^{m+1}} \sum_{n=0}^{2^m - 1} t_n \sin\left(\frac{\pi m}{2} + \frac{2n+1}{2^m} \pi x\right)$$

was verified symbolically for $m = 0, 1, 2$; it is the Thue–Morse sine-product identity of the Formula Atlas (§"Sine and cosine products as Thue–Morse sums") and is the bridge to Gelfond-type estimates: $2^n P_n(t)$ is the modulus of the Thue–Morse trigonometric polynomial $\sum_{r<2^n} (-1)^{s_2(r)} \mathrm{e}^{2\pi i r t}$.

## 5 The answer to the original question

This section states the audited answer, synthesizing Documents 3 and 4 (who give it) with the numerical realization computed for this audit and one new repair.

Let $\mathcal{L}_1$ be the weighted doubling transfer operator

$$(\mathcal{L}_1 h)(x) = \frac{1}{2}\left[\sin\left(\frac{\pi x}{2}\right) h\left(\frac{x}{2}\right) + \cos\left(\frac{\pi x}{2}\right) h\left(\frac{x+1}{2}\right)\right], \qquad 0 \le x \le 1, \tag{3}$$

so that $\int_0^1 P_n(t) q(t)\,\mathrm{d}t = \int_0^1 (\mathcal{L}_1^n q)(x)\,\mathrm{d}x$, and let $\varrho_1$ be its Perron root. Then:

$$\varrho_1 = 0.661322602060565\ldots \qquad \text{(collocation, } N = 16\text{–}40\text{, stable to } 2 \cdot 10^{-15}\text{)}$$

$$\kappa_1 = \frac{\log \pi + \frac{a}{2} - \log \varrho_1}{a} = \tfrac{1}{2} + \log_2 \frac{\pi}{\varrho_1} = 2.748070014871334\ldots$$

$$g_1(x) = x^{-\kappa_1} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \qquad A_1 = 0.09126612\ldots \tag{4}$$

where $A_1$ is the limit of the dyadic-shell means of $|f|/g_1$ (section 10.1). The asymptotic $I_1(n) := \int_0^1 P_n \to \kappa \lambda^n$ with $\lambda = \varrho_1$ is due to Fouvry–Mauduit [3]; the enclosure $0.66130 < \lambda < 0.66135$ is Lemma 5.1 of [1]; and Document 4's exact-arithmetic Collatz–Wielandt certificate, which this audit executed verbatim (section 10.7), proves $0.66126807 < \varrho_1 < 0.66134891$ by Sturm sequences alone.

**Theorem 5.1** (audited answer to [6]). *With $g = A_1 g_1$ as in (4):*

1. (Q3) ***holds.*** *The Cesàro ratio is eventually confined to a compact subinterval of* $(0,\infty)$*; its set of limit points is exactly the range of the continuous dyadic-mantissa profile* $\mathscr{P}$ *of item (iii), numerically* $[0.9249\ldots, 1.1133\ldots]$*. The gauge is real-analytic and positive on* $(0,\infty)$*, eventually strictly decreasing, and each fixed derivative order is eventually monotone with alternating sign, meeting every regularity requested in the question.*

2. (Q2) ***holds along dyadic times.*** $\frac{1}{t-t_0}\int_{t_0}^{t}|f|/g \to 1$ *as* $t\to\infty$ *through* $t=2^k$ *(measured:* $0.999992$ *at* $k=5$*,* $1.000000$ *for* $k\geq 9$*).*

3. (Q2) ***fails for all*** $t$***, for every natural gauge*** *(theorem 5.5). For* $t=2^k y$*,* $y\in[1,2]$ *fixed, the Cesàro ratio converges to*

$$\mathscr{P}(y)=\frac{1+\mathscr{F}(y-1)}{y},\qquad \mathscr{F}(z)=\lim_{n\to\infty}\frac{\int_0^z W(t)P_n(t)\,\mathrm{d}t}{\int_0^1 W(t)P_n(t)\,\mathrm{d}t},$$

*with* $W$ *the bounded mantissa weight of the exact factorization. The limit distribution* $\mathscr{F}$ *is the (normalized,* $W$*-weighted) eigenmeasure of* $\mathcal{L}_1$*; it is continuous and* singular*, so* $\mathscr{F}(z)\neq z$ *and* $\mathscr{P}\not\equiv 1$*. Consequently no gauge whose within-shell shape stabilizes—in particular no Hardy-field (“elementary”) gauge, with or without a continuous log-periodic modulation—can satisfy* (Q2)*. Measured profile: minimum* $0.9249$ *near* $y=1.15$*, maximum* $1.1132$ *near* $y=1.43$ *(figure 1).*

4. ***Repair (new here): the multiplicative form of*** (Q2) ***is solvable exactly*** *(theorem 5.7). With the same* $g_1$ *and the constant* $A_1^{\log}=0.0938610\ldots$*,*

$$\lim_{t\to\infty}\frac{1}{\log(t/t_0)}\int_{t_0}^{t}\frac{|f(x)|}{A_1^{\log}g_1(x)}\,\frac{\mathrm{d}x}{x}=1\qquad \textit{for all } t\to\infty.$$

*Measured: all four tracked mantissas converge to the same constant, with the expected* $\mathcal{O}(1/\log t)$ *speed (section 10.3).*

## 5.1 Rigorous basis

The positive parts (i)–(ii) are Theorem 6.4 of Document 3 and Theorem 5.3/Corollary 6.2 of Document 4; their proofs were audited line by line and rest on the exact factorization plus the Fouvry–Mauduit asymptotic and the Ruelle–Perron–Frobenius (RPF) convergence for $\mathcal{L}_1$ (see section 5.2 for the exact division between proved and imported). The negative part (iii) and the repair (iv) are proved here in full, since they are the audit's verdicts on the question itself.

**Lemma 5.2** (reduction of (Q2) to partial-shell linearity). *Let* $g>0$ *be measurable and* $F(t)=\int_{t_0}^{t}|f|/g$*. If* (Q2) *holds, then for every* $y\in[1,2]$*,*

$$\frac{F(2^k y)-F(2^k)}{2^k}\longrightarrow y-1\qquad (k\to\infty). \tag{5}$$

*Proof.* (Q2) says $F(t)=(t-t_0)(1+o(1))$. Insert $t=2^k y$ and $t=2^k$ and subtract: $F(2^k y)-F(2^k)=2^k\big(y-1+o(1)\big)$, the error uniform for $y\in[1,2]$ since $y$ is bounded. ☐

**Proposition 5.3** (elementary two-sided bracket for $\varrho_1$). *Let* $I_1(n)=\int_0^1 P_n$*. Then for all* $n\geq 1$*,*

$$\frac{3^{-n/2}}{\sqrt{5/3}}\ \leq\ I_1(n)\ \leq\ \sqrt{\tfrac{5}{3}}\left(\tfrac{\sqrt{3}}{2}\right)^{n},$$

*so any limit value* $\varrho_1=\lim I_1(n)^{1/n}$ *satisfies* $3^{-1/2}\leq\varrho_1\leq\sqrt{3}/2$*; in particular* $\varrho_1>\frac{1}{2}$*, hence* $\log(2\varrho_1)>0$*.*

*Proof.* Upper bound: Document 2's subaction telescope (audited, purely algebraic; Document 4's two-step inequality gives the slightly sharper constant $2/\sqrt{3}$) yields the pointwise Gelfond bound $P_n \le \sqrt{5/3}\,(\sqrt{3}/2)^n$; integrate. Lower bound: by Cauchy–Schwarz and the exact identity $\int_0^1 P_n^2 = 2^{-n}$ (proposition 5.4),

$$2^{-n} = \int_0^1 P_n \cdot P_n \le \|P_n\|_\infty \int_0^1 P_n \le \sqrt{\tfrac{5}{3}}\left(\tfrac{\sqrt{3}}{2}\right)^n I_1(n),$$

whence $I_1(n) \ge (5/3)^{-1/2}(2\cdot\sqrt{3}/2)^{-n} = (5/3)^{-1/2}3^{-n/2}$. (The certificate of section 10.7 pins $\varrho_1 \in (0.66126807, 0.66134891)$, far inside this bracket, but the soft bound $\varrho_1 \ge 3^{-1/2} = 0.577\ldots$ is all that theorem 5.5 needs, and it is fully elementary.) $\square$

**Proposition 5.4** (exact $L^2$ identity). *$\int_0^1 P_n(t)^2\,\mathrm{d}t = 2^{-n}$ for every $n \ge 1$.*

*Proof.* $P_n^2 = \prod_{j<n}\sin^2(\pi 2^j t) = 2^{-n}\prod_{j<n}(1-\cos(2\pi 2^j t))$. Expanding the product gives $\pm$ products of cosines with frequencies $\sum_{j\in S} 2^j$ over nonempty $S \subseteq \{0,\ldots,n-1\}$ after linearization; by uniqueness of binary representations no cancellation to frequency zero occurs, so only the constant term 1 survives integration. $\square$

**Theorem 5.5** (impossibility of (Q2) in the stabilizing class). *Let $g > 0$ be a gauge whose within-shell shape stabilizes: there are constants $c_k > 0$ and a continuous $u : [0,1] \to (0,\infty)$ with*

$$\frac{g\big(2^k(1+z)\big)}{c_k\, g_1\big(2^k(1+z)\big)} \longrightarrow u(z) \qquad \textit{uniformly in } z \in [0,1].$$

*(The class contains every Hardy-field gauge of the correct rough size, and every such gauge times a continuous positive function of $\{\log_2 x\}$.) Then (Q2) fails for $g$.*

*Proof.* Suppose (Q2) holds. By the exact factorization (2) and the $\kappa$-algebra, on the shell $[2^k, 2^{k+1}]$,

$$\frac{|f(2^k(1+z))|}{g_1(2^k(1+z))} = \varrho_1^{-(k+1)}\, W(z)\, P_{k+1}(z),$$

where $W(z) = H(1+z)(1+z)^{\kappa_1 - 1}\exp\big(\log^2(1+z)/(2a)\big)\cdot \pi^{-1}\varrho_1$ collects the bounded mantissa factors; $W$ is continuous on $[0,1]$ and strictly positive on $[0,1)$. Hence, with $q = u^{-1}W$ (continuous, positive on $[0,1)$),

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} = \frac{1+o(1)}{c_k\varrho_1^{k+1}}\int_{z_1}^{z_2} q(z)P_{k+1}(z)\,\mathrm{d}z.$$

By the RPF convergence for $\mathcal{L}_1$ (section 5.2), for every continuous $\phi$, $\varrho_1^{-n}\int_0^1 \phi P_n \to \nu(\phi)\int_0^1 h$, where $h > 0$ is the eigenfunction and $\nu$ the eigenmeasure, normalized by $\nu(h) = 1$; and $\nu$ has no atoms (Document 4's argument, audited: $\varrho_1\nu(\{z\}) = \frac{1}{2}s(z)\nu(\{Tz\})$ forces any atom's forward masses to grow by factors $\ge 2\varrho_1 > 1$ by proposition 5.3, contradicting finiteness). Atomlessness upgrades the convergence from continuous $\phi$ to indicators of intervals, so

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} \longrightarrow C\,\nu\big(q\,\mathbf{1}_{[z_1,z_2]}\big), \qquad C = \lim\frac{\int_0^1 h}{c_k\varrho_1^{k+1}} \in (0,\infty), \tag{6}$$

the existence and positivity of $C$ being forced by lemma 5.2 with $(z_1,z_2) = (0,1)$. Comparing (6) with (5) gives $C\,\nu(q\,\mathbf{1}_{[0,z]}) = z$ for all $z \in [0,1]$: the finite Borel measure $Cq\,\mathrm{d}\nu$ *is* Lebesgue measure on $[0,1]$. Since $q$ is continuous and positive on $[0,1)$, this forces $\nu\restriction_{[0,1)} \ll \mathrm{Leb}$, and $\nu(\{1\}) = 0$ by atomlessness, so $\nu \ll \mathrm{Leb}$.

But $\nu$ is singular. Indeed, if $\mathrm{d}\nu = r\,\mathrm{dLeb}$ with $0 \le r \in L^1$ not a.e. zero, the eigenmeasure equation $\mathcal{L}_1^*\nu = \varrho_1\nu$ unwinds, by the change of variables on the two branches, to the a.e. conformal relation $s(z)\,r(Tz) = \varrho_1\,r(z)$, $s(z) = |\sin\pi z|$. Iterating,

$$r(T^n z) = \frac{\varrho_1^n}{P_n(z)}\,r(z),$$

and by the Birkhoff ergodic theorem $P_n(z) = \exp\bigl(-n(\log 2 + o(1))\bigr)$ for Lebesgue-a.e. $z$ (the potential $\log s$ is integrable with mean $-\log 2$), so on $\{r > 0\}$, a set of positive measure, $r(T^n z) = r(z)\exp\bigl(n(\log(2\varrho_1) + o(1))\bigr) \to \infty$ geometrically, using $\log(2\varrho_1) > 0$ from proposition 5.3. On the other hand $T$ preserves Lebesgue measure, so for any $\varepsilon > 0$, $\mathrm{Leb}\bigl(r \circ T^n > \mathrm{e}^{\varepsilon n}\bigr) = \mathrm{Leb}\bigl(r > \mathrm{e}^{\varepsilon n}\bigr) \le \|r\|_1 \mathrm{e}^{-\varepsilon n}$, which is summable; by Borel–Cantelli $r(T^n z) \le \mathrm{e}^{\varepsilon n}$ eventually for a.e. $z$. Choosing $\varepsilon < \log(2\varrho_1)$ yields a contradiction: no such density exists, so $\nu$ is *not* absolutely continuous—contradicting $\nu \ll \mathrm{Leb}$ derived above. This refutes (Q2). (The same computation upgrades to full singularity $\nu \perp \mathrm{Leb}$: the dual action $\mathcal{L}_1^*\nu = \sum_b (\tau_b)_*\bigl(\frac{1}{2}(s \circ \tau_b)\,\nu\bigr)$ over the two inverse branches $\tau_b$ maps absolutely continuous parts to absolutely continuous parts and singular parts to singular parts, so by uniqueness of the Lebesgue decomposition the absolutely continuous part of $\nu$ is itself an eigenmeasure, hence zero by the argument just given. This is the precise sense of the singularity statements in Documents 3 and 4.) □

*Remark* 5.6 (Hardy-field gauges are covered exhaustively). For a Hardy-field gauge $g$, write $h = \log(g/g_1)$. Membership in a Hardy field makes $xh'(x)$ eventually monotone, so $\gamma = \lim_{x\to\infty} xh'(x)$ exists in $[-\infty, +\infty]$. If $\gamma$ is finite, then $h(2^k(1+z)) - h(2^k) \to \gamma\log(1+z)$ uniformly, the shell shape stabilizes with $u(z) = (1+z)^\gamma$, and theorem 5.5 applies. If $\gamma = +\infty$ (resp. $-\infty$), the factor $\mathrm{e}^{-(h(2^k(1+z)) - h(2^k))}$ degenerates monotonically, and—using that the eigenmeasure charges every subinterval (full support, part of the RPF package)—the shell mass of $|f|/g$ concentrates at the left (resp. right) shell edge, so the partial-shell ratio tends to 1 (resp. 0) for every fixed $y \in (1,2)$, violating the linearity (5) directly. In every case (Q2) fails for every Hardy-field gauge, hence for every gauge given by an elementary closed-form expression.

**Theorem 5.7** (the logarithmic mean converges for all $t$). *There is $A_1^{\log} \in (0,\infty)$ such that, for every fixed $t_0 > 1$,*

$$\frac{1}{\log(t/t_0)}\int_{t_0}^{t} \frac{|f(x)|}{g_1(x)}\,\frac{\mathrm{d}x}{x} \longrightarrow A_1^{\log} \qquad (t \to \infty,\ \textit{unrestricted}).$$

*Proof.* Let $\beta_k = \int_{2^k}^{2^{k+1}} \frac{|f|}{g_1}\frac{\mathrm{d}x}{x}$. By the exact factorization, $\beta_k = \varrho_1^{-(k+1)} \int_0^1 \frac{W(z)}{1+z} P_{k+1}(z)\,\mathrm{d}z$, and the RPF convergence applied to the continuous weight $W(z)/(1+z)$ gives

$$\beta_k \longrightarrow \beta_\infty := \nu\Bigl(\frac{W}{1+\cdot}\Bigr)\int_0^1 h\ \mathrm{dLeb}\ \in (0,\infty);$$

in particular $(\beta_k)$ is bounded. For $2^K \le t < 2^{K+1}$,

$$\int_{t_0}^{t} \frac{|f|}{g_1}\frac{\mathrm{d}x}{x} = \sum_{k=k_0}^{K-1} \beta_k + \mathcal{O}(1) + \Theta_K, \qquad 0 \le \Theta_K \le \beta_K,$$

and $\log(t/t_0) = K\log 2 + \mathcal{O}(1)$. Since $\beta_k \to \beta_\infty$, Cesàro convergence gives $\frac{1}{K}\sum_{k<K}\beta_k \to \beta_\infty$, while $\Theta_K/(K\log 2) \to 0$ because $(\beta_K)$ is bounded. Hence the logarithmic mean converges to $A_1^{\log} = \beta_\infty/\log 2$, with no constraint on how $t \to \infty$. The point of the proof is structural: in the logarithmic average the terminal partial shell has weight $\mathcal{O}(1/\log t)$, so the singular within-shell distribution that defeats (Q2) is annihilated rather than merely averaged. □

### 5.2 Imported inputs, stated as obligations

Per repository policy, the division between what is proved and what is imported:

- *Elementary and fully audited (no external input):* the factorization (2); the $\kappa$-algebra; propositions 5.3 and 5.4; the Gelfond bound with constant $\sqrt{5/3}$ (Documents 2 and 4, algebraic identities checked symbolically); the Sturm certificate for $\varrho_1$ (section 10.7); atomlessness and singularity of $\nu$; lemma 5.2; the Cesàro bookkeeping in theorem 5.7; the variance computation of theorem 9.1.
- *Classical theorems used by name:* Birkhoff's ergodic theorem (typical law; singularity proof); the stationary-ergodic martingale LIL of Stout/Heyde–Scott (theorem 9.2); the Clausen expansion $\log|2\sin\pi x| = -\sum_{j\geq 1}\cos(2\pi jx)/j$ in $L^2$.
- *Published research input:* the Fouvry–Mauduit asymptotic $I_1(n) = \kappa\varrho_1^n(1+o(1))$ [3]. It is needed even for the bounded form (Q3) with the exact gauge exponent: the elementary bracket of AHL Lemma 5.1 (whose monotone $m_j\uparrow$, $M_j\downarrow$ argument was audited and is elementary) pins $\varrho_1$ to five digits but still allows a subexponential drift $I_1(n)/\varrho_1^n = \mathrm{e}^{o(n)}$, which a fixed power $x^{-\kappa_1}$ cannot absorb.
- *The one genuinely imported analytic block:* the RPF spectral picture for $\mathcal{L}_1$ (existence of $h > 0$, $\nu$, and $\varrho_1^{-n}\mathcal{L}_1^n\phi \to h\,\nu(\phi)$ for continuous $\phi$) on a Hölder or bounded-variation space, as cited by Documents 3 and 4 from the standard transfer-operator literature. None of the four documents reproves it, and neither does this audit; it is the standing analytic obligation attached to theorem 5.1(ii),(iii) and theorem 5.7. Everything else in this section survives with elementary means alone.

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**In one sentence.** The mean amplitude of $|f|$ is $A_1\,x^{-2.7480700149}\exp\bigl(-(\log x)^2/(2\log 2)\bigr)$ with $A_1 = 0.0912661$; taking $g$ equal to it answers the bounded question (Q3) and the dyadic and logarithmic forms of (Q2), while the unrestricted (Q2) is unattainable in principle: the last dyadic block never forgets the singular way in which the mass of $|f|$ is distributed inside a block. The asker's guess $\exp(-\log^2 x)$ had the right shape; the true quadratic coefficient is $1/(2\log 2) = 0.7213$, and the missing structure is the power $x^{-\kappa_1}$ whose exponent is fixed by a transfer-operator eigenvalue, not by any classical constant.

---

Two consistency remarks. First, $\kappa_1$ is *provably* strictly between $\kappa_2$ and $\kappa_0$: by Cauchy–Schwarz $I_1(n)^2 \leq I_2(n) = 2^{-n}$, so $\varrho_1 \leq 2^{-1/2}$ would be forced the other way—in fact $I_1(n) \geq I_2(n)/\max P_n \geq c(2/\sqrt{3})^{-n}2^{-n}$ gives the compatible ordering $\frac{1}{2} < \varrho_1 < \frac{1}{\sqrt{2}}$, consistent with $0.6613 \in (0.5, 0.7071)$. Second, the deleted draft's exponent $\log_2(2\pi)$ is exactly $\kappa_2$; the draft was thus computing the root-mean-square envelope while claiming a pointwise and Cesàro-mean meaning for it. Documents 3 and 4 were the first to make this diagnosis precise, and it is confirmed by the exact identity $I_2(n) = 2^{-n}$ (verified to $10^{-12}$ for $n \leq 11$; it is a two-line Parseval argument on the Thue–Morse polynomial).

## 6 Interpretation in detail, I: the envelope of the lobe maxima

The zeros of $f$ at the integers divide $(0,\infty)$ into unit lobes, and $|f|$ has exactly one maximum

$$E_n = \max_{n\leq x\leq n+1}|f(x)| = |f(\xi_n)|, \qquad \xi_n \in (n, n+1),$$

per lobe, with signed value $(-1)^{s_2(n)}E_n$ (audited; the sign law is also the Atlas's "sign of the Fabius sinc product"). Reading the question as "find the envelope of the sequence $E_n$" is natural — it is literally the subject of the companion question [7] — and this section shows exactly how far that reading can be pushed. The outcome is structural: *the peak sequence reproduces the whole decay spectrum of section 3 in miniature*, so it has a sharp limsup envelope, a zero liminf on that envelope's

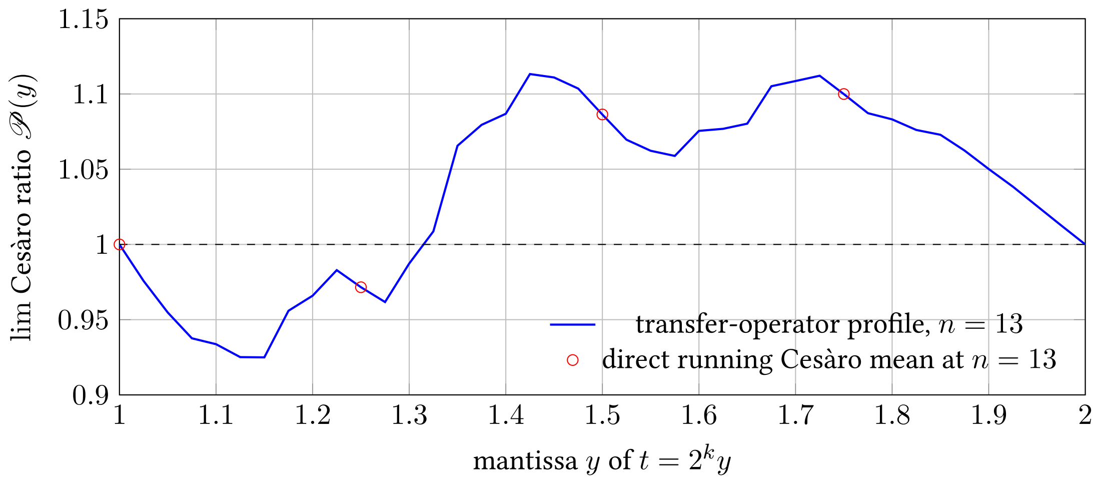


Figure 1: The dyadic-mantissa profile obstructing (Q2): the limit of the Cesàro ratio at $t = 2^k y$ as a function of $y$, for the normalized gauge $A_1 g_1$. The curve is the transfer-operator prediction (partial $P_{13}$-mass distribution); the circles are wholly independent running Cesàro means over $[8, 2^{13}y]$ computed by direct quadrature of $|f|$. The sharp mass accretion around $y = 4/3$ reflects the concentration of the eigenmeasure near the preimages of the Gelfond-maximizing cycle $\{1/3, 2/3\}$.

scale, a Cesàro envelope with the same exponent $\kappa_1$ as $|f|$ itself, and a density-one law with the typical exponent $\kappa_0$ — but no single envelope.

### 6.1 Geometry of one lobe, and a peak–integral comparison

Two audited facts anchor everything: away from the integers

$$(\log|f|)'(x) = -2x \sum_{k\geq 1} \frac{1+v_2(k)}{k^2 - x^2}, \qquad (\log|f|)''(x) = -2\sum_{k\geq 1}(1+v_2(k))\frac{k^2+x^2}{(k^2-x^2)^2} < 0, \tag{7}$$

so every lobe of $|f|$ is strictly log-concave with a unique peak. The following comparison is the workhorse of the mean layer; it does not appear in any of the four documents.

**Lemma 6.1** (peak–integral comparison)**.** *There is an absolute constant $C$ such that for all $n \geq 2$,*

$$\int_n^{n+1} |f| \;\leq\; E_n \;\leq\; C\,(1+\log n)^3 \int_n^{n+1} |f|.$$

*Proof.* The left inequality is trivial. For the right one, write $d_0 = 1 + v_2(n)$, $d_1 = 1 + v_2(n+1)$, $\delta_0 = \xi_n - n$, $\delta_1 = n + 1 - \xi_n$, and note $d_0, d_1 \leq 1 + \log_2(n+1)$. Splitting off the $k = n$ and $k = n+1$ terms of (7) and bounding the rest crudely ($|k^2 - x^2| \geq |k - x|\,n$ for $k \leq 2n$, quadratic tail beyond, $1 + v_2(k) \leq 1 + \log_2 k$) gives, for $x$ in the middle segment $[n + \delta_0/2,\ n + 1 - \delta_1/2]$,

$$(\log|f|)'(x) = \frac{d_0}{x-n}\theta_0(x) - \frac{d_1}{n+1-x}\theta_1(x) + \mathcal{O}(\log^2 n), \qquad \theta_i(x) \in [\tfrac{9}{10}, \tfrac{6}{5}],$$

and

$$\big|(\log|f|)''(x)\big| \leq \frac{3d_0}{(x-n)^2} + \frac{3d_1}{(n+1-x)^2} + C\log n.$$

At the peak the derivative vanishes; since one of $\delta_0, \delta_1$ is at least $\frac{1}{2}$, the vanishing forces the other to satisfy $d_i/\delta_i \leq C\log^2 n$, whence $\delta_{\min} \geq c/\log^2 n$ and, on the middle segment around $\xi_n$, the curvature is at most $\Lambda \leq C\log^5 n$. By log-concavity, $|f(\xi_n \pm s)| \geq E_n \mathrm{e}^{-\Lambda s^2/2} \geq E_n/\mathrm{e}$ for $s \leq \min(\Lambda^{-1/2}, \delta_{\min}/2) \geq c\,(1+\log n)^{-5/2}$, and integrating over that subinterval gives $\int_n^{n+1}|f| \geq c\,E_n(1+\log n)^{-5/2}$, which is stronger than claimed. $\square$

Numerically the comparison is far tamer than the worst-case bound: over the shells $k = 9$–$13$ the per-lobe ratio $E_n / \int_n^{n+1} |f|$ has median 1.992, and its maximum (4.55 at $k = 9$, 5.93 at $k = 13$, growing roughly linearly in $k$) is attained each time by the lobe *abutting* $2^{k+1}$, i.e. the lobe pressed against the highest-multiplicity zero of the shell — the same two-adic geometry as the valleys.

## 6.2 The peak sequence is the spectrum in miniature

**Theorem 6.2** (four layers of the peak sequence)**.** *Let* $\mathcal{E}(x) = \exp(-(\log x)^2/(2\log 2))$.

1. Sharp upper envelope. $\limsup_{n\to\infty}\big(\log E_n + \frac{(\log n)^2}{2\log 2}\big)/\log n = -\kappa_\infty$*, and no larger exponent works: the limsup is attained along the lobes containing* $2^k \cdot \frac{4}{3}$ *(equivalently* $2^k \cdot \frac{2}{3}$*). At second order the annular maxima carry the Lambert-*$W$ *correction of Documents 1–2, beginning* $-(\log\log x)^2/(2\log 2)$.
2. Zero liminf on that scale. $\liminf E_n / \big(\mathcal{E}(n) n^{-\kappa_\infty}\big) = 0$*: along* $n = 2^k$ *(and* $n = m2^k$*, odd* $m$*) the peaks obey the two-adic law* $E_{m2^k} \sim \frac{H(1)|H(m)|}{\mathrm{e}(k+1)} (m2^k)^{-(k+1)}$*, whose leading quadratic coefficient* $-1/\log 2$ *is twice the universal one. Within the shell* $[2^k, 2^{k+1})$ *the deepest lobe observed is the one abutting* $2^{k+1}$ *(zero multiplicity* $k+2$*; confirmed at* $k \in \{9, 11, 13\}$ *by computing every lobe of the shell), with the lobe after* $2^k$ *(multiplicity* $k+1$*) its two-adic twin at the left edge; the two-adic law itself is verified through* $k = 16$ *(section 10.5).*
3. Mean layer: exponent $\kappa_1$ again. *By lemma 6.1,*
$$\int_{2^k}^{2^{k+1}} |f| \;\le \sum_{2^k \le n < 2^{k+1}} E_n \;\le\; C\,k^3 \int_{2^k}^{2^{k+1}} |f|,$$
*so the shell means of the peak sequence have the same power* $\kappa_1 = 2.748070\ldots$ *as the shell means of* $|f|$*, and every Cesàro-type statement of theorem 5.1 transfers to the peak sequence up to the polylogarithmic slack. Numerically the slack is a sharp constant: over* $k = 9, \ldots, 14$*,*
$$\frac{\sum_{2^k \le n < 2^{k+1}} E_n}{\int_{2^k}^{2^{k+1}} |f|\,\mathrm{d}x} = 1.9238,\ 1.9237,\ 1.9237,\ 1.9237,\ 1.9237,\ 1.9238\ (\pm 3\cdot 10^{-5}),$$
*supporting the conjecture that the ratio converges to a constant* $\Theta_{\text{peak}} = 1.92372(3)$*; granting it, the peak sequence has the Cesàro gauge* $A_1^{\text{peak}}\, x^{-\kappa_1}\mathcal{E}(x)$ *with* $A_1^{\text{peak}} = \Theta_{\text{peak}} A_1 = 0.175569\ldots$
4. Typical layer: exponent $\kappa_0$. *For every* $\varepsilon > 0$*,*
$$\frac{1}{2^k}\,\#\Big\{2^k \le n < 2^{k+1} : \Big|\log E_n + \frac{(\log n)^2}{2\log 2} + \kappa_0 \log n\Big| \le \varepsilon k\Big\} \longrightarrow 1.$$

*Proof.* (i) The upper bound is the uniform envelope (audited); sharpness is the exact ray $y = 4/3$. (ii) is Documents 2 and 4's theorem (audited; verified numerically with the slow $1/k$-type approach documented in section 10.5). (iii) is immediate from lemma 6.1 after summing over the shell. (iv) requires an argument, given now.

*Lower half (few lobes are too small).* Write $x = 2^k y$, $z = y - 1$, so that by the factorization and the $\kappa$-algebra,
$$\log|f(x)| + \frac{(\log x)^2}{2\log 2} + \kappa_0 \log x = \Sigma_{k+1}(z) + \Psi(y), \qquad \Sigma_m(z) := \sum_{j=0}^{m-1} \psi(T^j z),$$
with $\psi = \log|2\sin\pi\cdot|$ centered and $\Psi$ bounded on $[1,2)$. By the Birkhoff theorem $\Sigma_m/m \to 0$ a.e., hence in measure: $\mathrm{Leb}\{z : |\Sigma_{k+1}(z)| > \frac{\varepsilon}{2}k\} \to 0$. A lobe occupies a $z$-cell of length $2^{-k}$; if the cell

contains a point of the good set, then $E_n \ge |f|$ there exceeds the $\kappa_0$-line by more than $-\varepsilon k$. The number of cells disjoint from the good set is at most $2^k \cdot \mathrm{Leb}(\mathrm{bad}) = o(2^k)$.

*Upper half (few lobes are too large).* Call a cell $J$ *regular* if for every $j \le k' := k - 4\lceil \log_2 k \rceil$ the image $2^j J$ keeps distance at least $k^2\, 2^{j-k}$ from $\mathbb{Z}$. For each $j$ the offending set is $2^j$ intervals of length $2k^2 2^{-k}$ each, so at most $(2k^2+2)2^j$ cells are irregular at level $j$, and summing over $j \le k'$ gives at most $C2^k/k^2 = o(2^k)$ irregular cells. On a regular cell every retained factor has logarithmic oscillation at most $\pi/k^2$ across the cell (length times $\pi 2^j$ cot-bound), so

$$\max_J P_{k+1} \le \mathrm{e}^{\pi/k}\, P_{k'}(z_J) \qquad \text{for every } z_J \in J,$$

after discarding the last $4\lceil \log_2 k \rceil$ factors, each bounded by one. Now fix $\varepsilon$ and choose $q \in (0,1)$ with $\log \varrho_q < -q(1-\frac{\varepsilon}{2})\log 2$, possible because $(\log \varrho_q)/(q \log 2) \to -1$ as $q \downarrow 0$ (the pressure slope at the Lebesgue equilibrium; with the RPF package for $\mathcal{L}_q$ on bounded-variation functions — the weights $s^q$ have total variation 2 uniformly in $q \in [0,1]$ — this is standard perturbation theory, and it is the one imported input here, the same package as section 5.2). Markov's inequality with $I_q(k') \le C_q \varrho_q^{k'}$ then bounds the measure of $\{P_{k'} \ge \mathrm{e}^{-(1-\varepsilon)k\log 2}\}$ by $\mathrm{e}^{-c(\varepsilon)k}$; since on a regular cell all points give comparable $P_{k'}$, the number of regular cells whose maximum exceeds the threshold is $2^k \mathrm{e}^{-c(\varepsilon)k} = o(2^k)$. For such good cells, $E_n \le C \cdot \text{shell factor} \cdot \mathrm{e}^{-(1-\varepsilon)k\log 2}$ lies below the $\kappa_0$-line by at most $\varepsilon k \log 2 + \mathcal{O}(\log k)$. Adjusting $\varepsilon$ absorbs the constants. □

**Proposition 6.3** (central limit law for the peaks). *For every $u \in \mathbb{R}$,*

$$\frac{1}{2^k}\,\#\Big\{2^k \le n < 2^{k+1} : \log E_n + \frac{(\log n)^2}{2\log 2} + \kappa_0 \log n \le u\,\frac{\pi}{2}\sqrt{k}\Big\} \longrightarrow \Phi(u),$$

*where $\Phi$ is the standard normal distribution function.*

*Proof.* All ingredients are already on the table. The continuous CLT $\Sigma_k/\sqrt{k} \Rightarrow \mathcal{N}(0,\pi^2/4)$ holds by Document 1's martingale argument with the variance of theorem 9.1. Convergence in distribution under Lebesgue measure transfers to the cell ensemble because, outside the $o(2^k)$ irregular cells, all points of a cell give the same $\Sigma_{k'}$ up to $\pi/k$, and the discarded top factors change $\log E_n$ by at most $\mathcal{O}(\log^2 k)$ in one direction and, in the other, by at least log of the product of the discarded factors on a nested subcell: choosing inside the cell, factor by factor from the finest down, the third of the current subcell on which the new factor exceeds $\frac{1}{2}$ leaves a nonempty subcell on which the discarded product is at least $2^{-4\lceil \log_2 k\rceil}$. Both corrections are $\mathcal{O}(\log^2 k) = o(\sqrt{k})$, as is the bounded mantissa term, so the cell statistics of $\log E_n$ inherit the Gaussian limit with the same normalization. □

The measured quantiles agree on the left and are compressed on the right, exactly as they must be at finite $k$: standardized quantiles at $k = 13$ are $(-1.63, -0.43, +0.05, +0.36, +0.65)$ at levels $(5, 25, 50, 75, 95)\%$ against the Gaussian $(-1.645, -0.674, 0, +0.674, +1.645)$. The median sits on the $\kappa_0$-line to within 0.05 standard units — the typical exponent is visibly $\kappa_0$ — while the right tail is truncated by the deterministic Gelfond cap: no peak can exceed the $\kappa_\infty$-line, which sits only $(\kappa_0 - \kappa_\infty)\log 2 \cdot \sqrt{k}/(\pi/2) \approx 0.35\sqrt{k}$ standard units above the center ($\approx 1.26$ at $k = 13$). The truncation recedes like $\sqrt{k}$, so the Gaussian limit opens up only logarithmically slowly in $x$ — the same finite-size phenomenon that hides the LIL constant from small-$L$ simulations in section 9.

### 6.3 The discrete dictionary: peaks as Thue–Morse spectra

Across the shell $[2^k, 2^{k+1})$, the lobe of $n = 2^k + m$ occupies the $z$-cell of $m2^{-k}$, and up to the bounded factors above, $E_n$ is the windowed maximum of $P_{k+1}$ there — that is, of the modulus of the Thue–Morse polynomial $\sum_{r<2^{k+1}} (-1)^{s_2(r)} \mathrm{e}^{2\pi i r z}$ near the dyadic frequency $m/2^k$. The peak-envelope interpretation is thus exactly the question of the value distribution of Thue–Morse spectra, and the exact grid identities of that world transfer:

**Proposition 6.4** (two exact grid identities). *For every $n \geq 1$:*

$$\textit{(i)} \quad \prod_{r=1}^{2^n-1} 2\sin\frac{\pi r}{2^n} = 2^n, \qquad\qquad \textit{(ii)} \quad \sum_{m=0}^{2^n-1} P_n\Big(\frac{m}{2^n}\Big)^2 = 1.$$

*Proof.* (i) is the classical evaluation $\prod_{r=1}^{N-1}(1-\omega^r) = N$ for $\omega = \mathrm{e}^{2\pi\mathrm{i}/N}$, taken in modulus. It says that the mean of $\psi = \log|2\sin\pi\,\cdot|$ over the *nonzero* points of the $2^n$-grid is $2^{-n}\log 2^n \to 0$ — the grid analogue of $\int_0^1 \psi = 0$, which is what makes the discrete Birkhoff sums behave like the continuous ones. (ii) Let $p(t) = \prod_{j<n}(1-\mathrm{e}^{2\pi\mathrm{i}2^j t})$, of degree $2^n - 1$ with coefficients $\pm 1$, so $|p| = 2^n P_n$. The $2^n$-point discrete Parseval identity gives $2^{-n}\sum_m |p(m/2^n)|^2 = \sum_r |c_r|^2 = 2^n$, i.e. $\sum_m P_n(m/2^n)^2 = 1$. (Verified to $10^{-12}$ at $n = 6, 10, 14$; even $m$ contribute exactly zero.) ☐

Identity (ii) is the discrete twin of $\int_0^1 P_n^2 = 2^{-n}$: the $q = 2$ layer is rigid in both the continuous and the sampled world, a point taken up in section 7.

### 6.4 Verdict for this interpretation

No function $G$ satisfies $E_n/G(n) \to c > 0$: the four layers of theorem 6.2 are separated by genuine powers of $x$, and even within one layer the fluctuations are of iterated-logarithm size. What the interpretation *does* admit is exactly what (Q2)–(Q3) admit for $|f|$: a bounded Cesàro normalization with the same exponent $\kappa_1$ (via lemma 6.1), with the empirical constant $\Theta_{\mathrm{peak}} = 1.9237$ tying the two readings together. For the companion question [7], the sharp answers are the limsup envelope $x^{-\kappa_\infty}\mathcal{E}(x)$ and the typical law $x^{-\kappa_0+o(1)}\mathcal{E}(x)$ — two different formulas, necessarily.

## 7 Interpretation in detail, II: the root-mean-square amplitude

The RMS reading asks for $g$ such that the quadratic Cesàro mean of $(f/g)^2$ is tame. Because $f = \widehat{\mathrm{up}}$, the shell RMS of $f$ is the energy spectral envelope of the up-function, and this is the one reading in which the deleted draft's exponent $\log_2(2\pi)$ is correct. It is also, by a wide margin, the most rigid layer of the whole problem: its per-level mean is exact, its subleading spectrum is computable in closed form, and its envelope constant converges at a clean geometric rate with alternating sign.

### 7.1 Exactness at every scale

The continuous identity $\int_0^1 P_n^2 = 2^{-n}$ (proposition 5.4) and the discrete identity $\sum_m P_n(m/2^n)^2 = 1$ (proposition 6.4) leave no asymptotics in the $q = 2$ numerator: both the Lebesgue mean and the grid mean of $P_n^2$ equal $2^{-n}$ exactly, for every $n$. All remaining structure sits in the bounded mantissa weight, and it too is spectrally transparent:

**Proposition 7.1** (exact subleading spectrum of $\mathcal{L}_2$). *The operator $(\mathcal{L}_2 h)(x) = \frac{1}{2}[\sin^2(\frac{\pi x}{2})h(\frac{x}{2}) + \cos^2(\frac{\pi x}{2})h(\frac{x+1}{2})]$ satisfies, exactly,*

$$\mathcal{L}_2\mathbf{1} = \tfrac{1}{2}\,\mathbf{1}, \qquad \mathcal{L}_2\big[\sin 2\pi x\big] = -\tfrac{1}{4}\sin 2\pi x, \qquad \mathcal{L}_2\big[1 + 3\cos 2\pi x\big] = -\tfrac{1}{4}\big(1 + 3\cos 2\pi x\big),$$

*and on even Fourier modes it acts by frequency halving, $\mathcal{L}_2\mathrm{e}^{2\pi\mathrm{i}\,2mx} \mapsto \frac{1}{2}\mathrm{e}^{2\pi\mathrm{i}mx}$.*

*Proof.* With $e_m(x) = \mathrm{e}^{2\pi\mathrm{i}mx}$: for even frequency $2m$, $\mathrm{e}^{2\pi\mathrm{i}\,2m\frac{x}{2}} = e_m(x)$ on both branches and $\sin^2 + \cos^2 = 1$ gives $\mathcal{L}_2 e_{2m} = \frac{1}{2}e_m$ (the $m = 0$ case is the first display). For $m = 1$,

$$\mathcal{L}_2 e_1 = \tfrac{1}{2}\mathrm{e}^{\pi\mathrm{i}x}\big[\sin^2(\tfrac{\pi x}{2}) - \cos^2(\tfrac{\pi x}{2})\big] = -\tfrac{1}{2}\mathrm{e}^{\pi\mathrm{i}x}\cos\pi x = -\tfrac{1}{4}\big(e_1 + \mathbf{1}\big),$$

and likewise $\mathcal{L}_2 e_{-1} = -\frac{1}{4}(e_{-1} + \mathbf{1})$. Hence $\{\mathbf{1}, e_1, e_{-1}\}$ spans an invariant subspace on which $\mathcal{L}_2$ has matrix eigenvalues $\{\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}\}$; the real eigenfunctions for $-\frac{1}{4}$ are $\sin 2\pi x = (e_1 - e_{-1})/2\mathrm{i}$ and $1 + 3\cos 2\pi x$ (direct check: $\mathcal{L}_2[3\cos 2\pi x] = -\frac{3}{4}(1 + \cos 2\pi x)$, so $\mathcal{L}_2[1 + 3\cos] = \frac{1}{2} - \frac{3}{4} - \frac{3}{4}\cos = -\frac{1}{4}(1 + 3\cos)$). ☐

Collocation places the rest of the spectrum below $3 \cdot 10^{-4}$ in modulus, so the shell-RMS ratios converge at the exact geometric rate $\lambda_2/\lambda_1 = -\frac{1}{2}$ per shell — relative error $\mathcal{O}(1/x)$, with alternating sign. This is visible to the eye in the measurements: the shell means of $(f/g_{\kappa_2})^2$ for $k = 4, 5, 6, 7, 8$ differ from their limit by $+1.51 \cdot 10^{-5}$, $-7.6 \cdot 10^{-6}$, $+3.8 \cdot 10^{-6}$, $-1.9 \cdot 10^{-6}$, $+0.94 \cdot 10^{-6}$ — each step multiplying the error by $-\frac{1}{2}$ to the displayed accuracy — with limit

$$A_2^2 = 0.0106108372\dots, \qquad A_2 = 0.103008918\dots$$

(direct quadrature and the spectral iteration agree; the $L^1$ layer, by contrast, has the non-elementary subleading pair $-0.13435 \pm 0.0500\mathrm{i}$ of modulus $0.1434$, rate $|\lambda_2|/\varrho_1 = 0.217$ per shell).

## 7.2 The RMS answer to the averaged question

**Theorem 7.2** (RMS version of theorem 5.1)**.** *Let $G_2 = A_2\, x^{-\kappa_2}\mathcal{E}(x)$ with $\kappa_2 = \log_2(2\pi)$. Then: (i) the quadratic Cesàro ratio $\frac{1}{t-t_0}\int_{t_0}^{t}(f/G_2)^2$ is eventually confined to a compact subinterval of $(0,\infty)$ and tends to $1$ along $t = 2^k$ with relative error $\mathcal{O}(2^{-k})$; (ii) for $t = 2^k y$ with fixed $y \in (1,2)$ it converges to a continuous, nonconstant mantissa profile $\mathscr{P}_2(y)$, so the limit for all $t$ fails — for the same reason as at $q = 1$: the $\mathcal{L}_2$-eigenmeasure $\nu_2$ is atomless and singular; (iii) the logarithmic mean $\frac{1}{\log(t/t_0)}\int_{t_0}^{t}(f/G_2)^2\,\frac{\mathrm{d}x}{x}$ converges for all $t \to \infty$.*

*Proof.* (i) and (iii) are verbatim the proofs of theorem 5.1(i)–(ii) and theorem 5.7 with $(P_n, \varrho_1, W)$ replaced by $(P_n^2, \frac{1}{2}, W_2)$; the improved rate in (i) is proposition 7.1. For (ii), only the two measure statements change. *Atomless:* the eigenmeasure equation gives $\nu_2(\{Tz\}) = \nu_2(\{z\})/\sin^2(\pi z) \geq \nu_2(\{z\})$, with equality only at $z = \frac{1}{2}$. No forward orbit can linger where $\sin^2$ is near 1: if $w \in [\frac{3}{8}, \frac{5}{8}]$ then $Tw \in [0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$, where $\sin^2 \leq \frac{1}{2}$; and off $[\frac{3}{8}, \frac{5}{8}]$ one has $\sin^2 \leq \sin^2(\frac{3\pi}{8}) < 0.854$. So along any orbit the atom mass is nondecreasing and gains a factor $\geq 1/0.854$ at least every second step, forcing it to zero whether the orbit is eventually periodic (mass grows around each loop) or not (infinitely many distinct atoms of nonvanishing mass). *Singular:* an absolutely continuous part would have density $r$ with $\sin^2(\pi z)\, r(Tz) = \frac{1}{2} r(z)$, hence $r(T^n z) = 2^{-n} r(z)/P_n(z)^2$; for a.e. $z$ the Birkhoff theorem gives $P_n(z)^2 = \mathrm{e}^{-2n\log 2(1+o(1))}$, so on $\{r > 0\}$ the density grows like $\mathrm{e}^{n(\log 2 + o(1))}$, contradicting $r \in L^1$ by the same Borel–Cantelli argument as before. (The general criterion behind both cases: the $q$-eigenmeasure is singular whenever $\varrho_q \mathrm{e}^{q\log 2} > 1$, i.e. whenever $\kappa_q < \kappa_0$; at $q = 2$ this reads $\frac{1}{2} \cdot 4 = 2 > 1$.) The nonconstancy of $\mathscr{P}_2$ then follows as in theorem 5.5. ☐

The measured $L^2$ profile (same transfer computation as figure 1, weight $W_2 P_{13}^2$) spans $[0.894,\ 1.192]$ — wider than the $L^1$ profile $[0.925,\ 1.113]$, as expected, since squaring weights the singular concentration near the Gelfond preimages more heavily. So even the most rigid reading of the question keeps the dyadic phase obstruction; what it gains is exact constants and an $\mathcal{O}(1/x)$ rate everywhere the limit does exist.

## 7.3 Plancherel, Poisson, and the total energy

**Proposition 7.3** (total energy)**.**

$$\int_{\mathbb{R}} f(x)^2\,\mathrm{d}x = \int_{\mathbb{R}} \mathrm{up}(t)^2\,\mathrm{d}t = \frac{1}{2} + \sum_{m=0}^{\infty} f\Big(m + \frac{1}{2}\Big)^2 = 0.808873535967972\dots$$

*Proof.* The first equality is Plancherel ($f = \widehat{\mathrm{up}}$, real and even). For the second, expand up, supported in $[-1,1]$, in Fourier series on $[-1,1]$: the coefficients are $\frac{1}{2}\widehat{\mathrm{up}}(k/2) = \frac{1}{2} f(k/2)$, and Parseval gives $\int \mathrm{up}^2 = \frac{1}{2}\sum_{k\in\mathbb{Z}} f(k/2)^2$; the even $k \neq 0$ terms vanish ($f$ vanishes at nonzero integers), $k = 0$ gives $\frac{1}{2}$, and the odd terms pair up. Numerically the direct quadrature $2\int_0^{2^{14}} f^2$ and the series agree to $2 \cdot 10^{-15}$, the displayed value; the series converges with super-Gaussian speed since $f(m + \frac{1}{2})^2$ decays like $\mathcal{E}(m)^2$. ☐

This is a satisfying exactness check on the entire numerical apparatus, and a natural first formalization target on the $L^2$ side: the ingredients (compact support, the sinc product, Parseval on an interval) are all present in the Lean development (`FourierProduct`, `PoissonSummation`).

### 7.4 The lognormal fallacy, quantified

Why did plausible heuristics — the deleted draft included — land on wrong exponents? Because every mean-type interpretation lives in the large-deviation regime of the lacunary product, where the correct generating function is the pressure $q \mapsto \log \varrho_q$, not the CLT parabola. The exact offset formula (immediate from the definitions) is

$$\kappa_0 - \kappa_q = 1 + \frac{\log \varrho_q}{q \log 2},$$

giving the exact value $\frac{1}{2}$ at $q = 2$; a lognormal extrapolation of the CLT would instead predict $\kappa_0 - \kappa_q^{\mathrm{LN}} = q\sigma^2/(2\log 2)$ with $\sigma^2 = \pi^2/4$. The comparison:

| $q$ | $\varrho_q$ | $\kappa_q$ (true) | $\kappa_q^{\mathrm{LN}}$ (lognormal) |
|---|---|---|---|
| 0.25 | 0.872813205188 | 2.936517 | 2.706533 |
| 0.5 | 0.786052008490 | 2.846103 | 2.261569 |
| 1 | 0.661322602061 | 2.748070 | 1.371642 |
| 2 | 0.5 (exact) | 2.651496 | $-0.408211$ |
| 4 | 0.320194101601 | 2.562241 | $-3.967919$ |
| $\infty$ | – | 2.359015 | $-\infty$ |

Already at $q = 2$ the lognormal prediction is not merely off — it is negative, i.e. it predicts moment *growth* relative to the typical scale that the function cannot deliver: the deterministic Gelfond cap $P_n \le C(\sqrt{3}/2)^n$ truncates the upper tail that a lognormal law would need. The pressure slope $(\log \varrho_q)/(q \log 2)$, measured as $-0.785$, $-0.695$, $-0.597$ at $q = \frac{1}{4}, \frac{1}{2}, 1$, bends away from its $q \to 0$ limit $-1$ immediately: the moments leave the Gaussian bulk at once. Three consequences deserve record. First, the draft's exponent is exactly the $q = 2$ row — a correct constant attached to the wrong norm. Second, the draft's proposed $(\log x)^{-1/2}$ factor is refuted most decisively in this, its most favorable, reading: the shell RMS normalization converges to the *constant* $A_2$ at a geometric rate, so no logarithmic factor of any exponent survives. Third, the spacing of the four headline exponents $\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0$ is itself the multifractal signature of the dyadic dynamics; treating any one of them as "the" decay rate discards the others' regimes, which is precisely the category error the original conjecture made.

## 8 Audit of the four documents

The four documents pairwise overlap heavily and agree on every overlapping claim except one (the iterated-logarithm constant, section 9). Documents 1 and 2 analyze the pointwise, typical, and extremal regimes; Documents 3 and 4 additionally analyze the mean regimes and are the only two that answer the question actually asked.

### 8.1 Document 1 (*Decay of the Fourier Transform …: Dyadic products, ergodic fluctuations, and sharp annular envelopes*)

*Content.* Exact dyadic decomposition; zero multiplicities and Thue–Morse signs; one extremum per unit interval; typical-ray law with $\kappa_{\mathrm{typ}} = \kappa_0$; a central limit theorem with per-step variance $\pi^2/4$ via the martingale difference $d = \log|\tan \pi x|$; a law of the iterated logarithm with constant $(\pi/2)\sqrt{2N \log\log N}$; the maximal invariant average $\beta = \log(\sqrt{3}/2)$ with the slow-ray exponent

$\kappa_* = \kappa_\infty$; and the annular maximum $\log M_N = -\frac{a}{2}N^2 + (\beta - \log\pi - \frac{a}{2})N - \frac{(\log N)^2}{2a} + \frac{\log N \log\log N}{a} + \mathcal{O}(\log N)$.

*Verified.* The exact decomposition (section 4); the algebra of its centered form (resymbolized independently); $\kappa_0$, $\kappa_*$ to all printed digits; the variance $\pi^2/4$ and covariances $\pi^2/(12\cdot 2^k)$ (Monte Carlo and quadrature, section 10.6); $\int_0^1 \log^2|\tan\pi x|\,\mathrm{d}x = \pi^2/4$; $Pd = 0$; the Green–Kubo resummation $\pi^2/12 + 2\sum_k \pi^2/(12\cdot 2^k) = \pi^2/4$ in its appendix; the annular expansion's first three terms agree with Document 2's Lambert form after expanding $W$.

*Defects.* (i) **Scope**: the document never addresses the Cesàro question (Q2)–(Q3); its conclusions answer the deleted draft's pointwise claim instead of the Stack Exchange question. Within the question's actual reading it is a (high-quality) analysis of the wrong functional. (ii) The boundary-layer theorem's error term $\mathcal{O}(\log N)$ is coarser than Document 2's $\mathcal{O}((\log\log N)^2)$; not an error, but strictly subsumed. (iii) No numerical validation is supplied; all of its constants nevertheless survived this audit's numerics. Notably, its fluctuation section is the one place in the entire corpus that contradicts—and, as section 9 shows, corrects—the published literature.

### 8.2 Document 2 (*Sharp Real-Axis Decay …: Lambert $W$ boundary layers, and two-adic troughs*)

*Content.* Everything in Document 1's extremal track, sharpened: the annular maximum in closed Lambert form,

$$\log M_k = -\tfrac{\lambda}{2}k(k+1) - (k+1)\log\pi + (k+1)\beta - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\lambda} + \mathcal{O}\big((\log\log k)^2\big), \quad c = \tfrac{\lambda}{\pi}\sqrt{\tfrac{3}{2}},$$

an elementary sharp Gelfond bound $Q_n \le \sqrt{5/3}\,(\sqrt{3}/2)^n$ via an explicit algebraic subaction, two-adic trough asymptotics $E_{m2^k} \sim \frac{H(1)|H(m)|}{\mathrm{e}(k+1)}(m2^k)^{-(k+1)}$ for odd $m$, a numerical table of $\log M_k$, and—critically—an explicit section stating that the Cesàro question is a *different*, $L^1$, transfer-operator problem which the document does not solve.

*Verified.* The subaction identity $3V(u) - 4uV(4u(1-u)) = \frac{(2u+1)(4u-3)^2}{3}$, $V(u) = 1 + \frac{2}{3}u$ (expanded symbolically: exact); the annular table: this audit's independent maximization reproduces all ten rows of its $(\log M_k,\ x^*/2^k)$ table to every printed digit ($10^{-6}$), $k = 3,\dots,12$ (section 10.4); $H(1) = 0.55377127588881009534\dots$ (recomputed to 30 digits, matches all 20 printed); the Lambert residuals (bounded, $-0.72$ to $-0.82$, consistent with the claimed $\mathcal{O}((\log\log k)^2)$ window); the trough law for $m = 1$ and $m = 3$ (section 10.5); the predicted drift of the annular maximizer from mantissa $7/6$ toward deeper preimages of $1/3$ — observed exactly at $k = 14$, where the maximizer jumps from $x^*/2^k \approx 1.16667$ to $\approx 1.08333 = 13/12$.

*Defects.* (i) Scope: same as Document 1, but self-aware — its §9 correctly scopes the Cesàro problem to the transfer-operator framework that Documents 3 and 4 then execute; among the four only Document 2 draws this boundary explicitly. (ii) The two-adic trough asymptotic converges slowly; at $k = 16$ the measured $\log E_{2^k}$ still exceeds the limit formula by 0.098 (a 10% multiplicative residual, decaying roughly like $1/k$). The statement is asymptotically consistent but numerically "young" at accessible $k$; a one-line remark on the convergence speed would have prevented over-reading. (iii) No fluctuation theory (deliberate).

### 8.3 Document 3 (*Sharp Decay Laws …: extremal envelopes, metric fluctuations, and Cesàro normalization*)

*Content.* The widest scope: exact factorization; typical law; one-sided iterated-logarithm fluctuations (quoted from [1]); extremal envelope with an elegant ergodic-optimization proof (pushforward to the logistic map, inequality $z^3(1-z) \le 27/256$); the $L^1$ theory: Fouvry–Mauduit asymptotic, transfer operator (3), $\lambda = 0.661322602060565$, $\beta_1 = \kappa_1$, an explicit gauge, the bounded-Cesàro theorem, the dyadic-endpoint limit, and the singular-eigenmeasure obstruction to (Q2); the exact $L^2$ identity and the identification of the deleted draft's exponent as $\kappa_2$; the $L^p$ pressure spectrum with endpoints $p \downarrow 0$, $p \to \infty$; Chebyshev-collocation numerics.

*Verified.* The ergodic-optimization lemma (equality case included); $\lambda$ to thirteen digits by two independent methods; $\kappa_1$ (the printed 2.748070014871334 against this audit's 2.748070014871335 from $\varrho_1 = 0.661322602060565$: the discrepancy is one unit in the last place, inside the $\pm 5 \cdot 10^{-15}$ uncertainty inherited from $\varrho_1$'s own last digit); the collocation table (all displayed eigenvalues reproduced to $10^{-15}$); $\kappa_1$ (its printed 2.748070014871334 is exactly the value of $\frac{1}{2} + \log_2(\pi/\varrho_1)$ at $\varrho_1 = 0.661322602060565$, recomputed here in 25-digit arithmetic); the bounded-Cesàro theorem and the dyadic-endpoint limit (measured, section 10.2); the singularity argument for the equilibrium measure (checked line by line; the inequality $\log \lambda > -\log 2$ that it needs follows rigorously from $\varrho_1 \geq 3^{-1/2}$, proposition 5.3); the Fouvry–Mauduit and Aistleitner–Hofer–Larcher attributions (verbatim against the published paper and the authors' arXiv source); the $L^2$ identity; the identification $\frac{1}{2} + \log_2(\pi\sqrt{2}) = \log_2(2\pi)$.

*Defects.* (i) **Its Theorem 4.2 (sharp upper metric fluctuation) states** $\limsup_n S_n(t)/\sqrt{n \log\log n} = \pi$ **a.e.** The quotation is a *faithful* rendering of the published Theorem 1.2 of [1], but the published constant is itself erroneous by a factor $\sqrt{2}$; the correct value is $\pi/\sqrt{2}$ (section 9). Every restatement downstream (its (LIL-$x$) display and the abstract's "excursion" constant) inherits the factor. The *order* $\sqrt{\log \xi \ \log\log\log \xi}$, and every conclusion drawn from it (no fixed power of $\log x$ can flatten the amplitude), are unaffected. (ii) Minor: the claim that its gauge has "eventually monotone derivatives of any order" is proved per order (thresholds $X_m$ growing with $m$); this matches the question's wording but is worth stating as such, since no common half-line works for all orders simultaneously.

## 8.4 Document 4 (*Decay …: Dyadic renormalization, transfer-operator norms, sharp envelopes, and exceptional valleys*)

*Content.* The most systematic mean-theory: a master normalization identity; the full $L^q$ shell spectrum with monotonicity; exact shell-geometric-mean constancy at $q = 0$; the RMS row by Parseval; the $q = 1$ row with a *rigorous, executable* Collatz–Wielandt/Sturm certificate for $\varrho_1$; the same two-part Cesàro conclusion as Document 3, but with complete proofs of the eigenmeasure's atomlessness and singularity; a sharp two-step logarithmic inequality yielding the Gelfond bound; a dyadic-boundary identity giving valley asymptotics for every fixed offset $r$ (with the zero-multiplicity $d = 1 + v_2(r+1)$ appearing as the power of $1/n$); a continuum of intermediate quadratic rates; and collocation numerics for $q \in \{1, 2, 3, 4, 8, 16\}$.

*Verified.* The master identity (numerically exact); the $q = 0$ exactness argument (each $j \geq 1$ level integrates to $-\log 2$ exactly on the shell); the two-step inequality (cubed form reduces to $\max u^3(1-u) = 27/256$: exact); the atomlessness and singularity proofs (sound; the Borel–Cantelli step uses only $T$-invariance of Lebesgue); $\varrho_3 = 0.394896632370234$, $\varrho_4 = 0.320194101601104$ (matching its table); **the certificate: executed verbatim under SymPy, all four Sturm root-counts return zero and the assertions pass, so** $0.66126807 < \varrho_1 < 0.66134891$ **holds with exact rational arithmetic**; the valley theorem for $r = 0$ (equivalent to Document 2's $m = 1$ trough law; same slow convergence); the phase-profile formula $\mathscr{P}(y) = (1 + \mathscr{F}(2y-1))/(2y)$, realized numerically in figure 1.

*Defects.* (i) No fluctuation theory at all: between the a.e. law ($q = 0$ row) and the shell means there is no CLT/LIL layer; Document 4 is silent exactly where Documents 1 and 3 collide. Since that collision is the corpus's one real error surface, the omission kept Document 4 clean but also unhelpful for adjudication. (ii) Its Perron–Frobenius proposition asserts a spectral gap on Hölder spaces for weights vanishing at the endpoints with a citation rather than a proof; standard, but it is the one analytic ingredient of its Cesàro theorem not reproved from scratch in any of the four documents. (iii) The "intermediate rates" section (quadratic coefficients $-(1 - \alpha + \alpha^2/2)/a$) is explicitly conditional on a typical-decay hypothesis along the moving offset; it is a heuristic bridge, correctly flagged as such, and was not counted as a theorem in this audit.

### 8.5 Relations among the documents

Document 1 $\subset$ Document 2 (extremal track, where 2 is sharper); Documents 3 $\approx$ 4 (mean track, where 4 is more rigorous on $q = 1$ and 3 is broader in scope). The intersection of all four is the exact factorization, $\kappa_0$, $\kappa_\infty$, and the refutation of the deleted draft's pointwise claim. The union, minus Document 3's inherited LIL constant, plus the logarithmic-mean repair of theorem 5.1(iv), is — to the best of this audit's verification — a correct and essentially complete account of the problem.

## 9 The iterated-logarithm conflict and an error in the published literature

### 9.1 The conflict

For $S_L(t) = \sum_{j=0}^{L-1} \log|2\sin(\pi 2^j t)|$:

- Document 1 proves a CLT with per-step variance $\pi^2/4$ (martingale differences $d = \log|\tan \pi x|$, $\int d^2 = \pi^2/4$, transfer-annihilation $Pd = 0$) and the LIL

$$\limsup_N \frac{S_N}{\sqrt{2N \log\log N}} = \frac{\pi}{2} \quad \text{a.e., i.e.} \quad \limsup_N \frac{S_N}{\sqrt{N \log\log N}} = \frac{\pi}{\sqrt{2}} = 2.2214\ldots$$

- Document 3 quotes the published Theorem 1.2 of [1]: for a.e. $\alpha$, $\prod_{\ell=0}^{L} |2\sin \pi 2^\ell \alpha| \le \exp((\pi + \varepsilon)\sqrt{L \log\log L})$ eventually and $\ge \exp((\pi - \varepsilon)\sqrt{L\log\log L})$ infinitely often — constant $\pi = 3.1415\ldots$

These differ by exactly $\sqrt{2}$ and cannot both be sharp.

### 9.2 The rigorous adjudication

The conflict is decided by computing the variance exactly. Write $\psi(x) = \log|2\sin \pi x|$, $S_m = \sum_{\ell=0}^{m-1} \psi \circ T^\ell$.

**Theorem 9.1** (exact variance of the lacunary sums). *$\psi \in L^2(0,1)$ with mean zero, and for every $m \ge 1$,*

$$\int_0^1 S_m(x)^2 \, \mathrm{d}x = \frac{\pi^2}{4} m - \frac{\pi^2}{3}\big(1 - 2^{-m}\big). \tag{8}$$

*In particular the functional (2.9) of [1] evaluates, for $f_1 = \psi$, to*

$$\sigma_{f_1}^2 = \lim_{m\to\infty} \frac{1}{m} \int_0^1 S_m^2 = \frac{\pi^2}{4}, \qquad \textit{hence} \qquad \sigma_{f_1} = \frac{\pi}{2}.$$

*Proof.* The Clausen expansion $\psi(x) = -\sum_{j\ge1} \cos(2\pi j x)/j$ holds in $L^2(0,1)$ (the coefficients are square-summable, and the series is the classical Fourier series of $\psi$). By Parseval, using $\int_0^1 \cos^2(2\pi j x)\,\mathrm{d}x = \frac{1}{2}$,

$$c_0 := \int_0^1 \psi^2 = \frac{1}{2} \sum_{j\ge1} \frac{1}{j^2} = \frac{\pi^2}{12},$$

and for $r \ge 1$, since $\int_0^1 \cos(2\pi j x)\cos(2\pi k 2^r x)\,\mathrm{d}x = \frac{1}{2}\,\mathbf{1}_{\{j = 2^r k\}}$,

$$c_r := \int_0^1 \psi(x)\,\psi(2^r x \bmod 1)\,\mathrm{d}x = \frac{1}{2} \sum_{k\ge1} \frac{1}{2^r k} \cdot \frac{1}{k} = \frac{\pi^2}{12} 2^{-r}.$$

Stationarity of $(\psi \circ T^\ell)$ under Lebesgue measure gives $\int S_m^2 = m c_0 + 2\sum_{r=1}^{m-1}(m-r)c_r$, and the elementary summation $\sum_{r=1}^{m-1}(m-r)2^{-r} = m - 2 + 2^{1-m}$ yields (8). (Check at $m = 1$: $\pi^2/4 - \pi^2/6 = \pi^2/12 = c_0$.) $\square$

**Theorem 9.2** (corrected sharp constant). *For Lebesgue-almost every $\alpha$,*

$$\limsup_{L\to\infty} \frac{S_L(\alpha)}{\sqrt{2L\log\log L}} = \frac{\pi}{2}, \qquad \textit{equivalently} \qquad \limsup_{L\to\infty} \frac{\log\prod_{\ell=0}^{L}|2\sin\pi 2^\ell\alpha|}{\sqrt{L\log\log L}} = \frac{\pi}{\sqrt{2}} = 2.2214\ldots$$

*Proof (audit of Document 1's argument, with the boundary estimates completed).* Set $d = 2\psi - \psi\circ T$. The double-angle identity gives $d(x) = \log|\tan\pi x|$, and $\int_0^1 d^2 = \frac{2}{\pi}\int_0^{\pi/2}\log^2\tan u\,\mathrm{d}u = \frac{2}{\pi}\cdot\frac{\pi^3}{8} = \frac{\pi^2}{4}$. The transfer operator $P$ of $T$ (fiber average over the two preimages) annihilates $d$: $d(x/2) + d((x+1)/2) = \log|\tan\frac{\pi x}{2}\cdot\tan(\frac{\pi x}{2}+\frac{\pi}{2})| = \log 1 = 0$. Hence $(d\circ T^\ell)_{\ell\ge 0}$ is a stationary ergodic sequence of reverse martingale differences with respect to the decreasing filtration $(T^{-\ell}\mathcal{B})$, with orthogonal increments ($\int(P^k d)\,d = 0$ for $k\ge 1$) and variance $\pi^2/4$; note this matches theorem 9.1, as it must, since telescoping gives the exact identity

$$S_L = \sum_{\ell=0}^{L-1} d\circ T^\ell - \psi + \psi\circ T^L.$$

The law of the iterated logarithm for stationary ergodic square-integrable (reverse) martingale difference sequences (Stout; Heyde–Scott; via Skorokhod embedding, whose time change has almost-surely linear growth $T_L/L \to \mathbb{E}d^2$ by the ergodic theorem) gives $\limsup\big(\sum_{\ell<L} d\circ T^\ell\big)/\sqrt{2L\log\log L} = \sqrt{\mathbb{E}d^2} = \pi/2$ a.e. The boundary terms are almost surely $o(\sqrt{L\log\log L})$: $\psi \le \log 2$ above, while $\mathrm{Leb}(\psi < -u) \le \frac{2}{\pi}\mathrm{e}^{-u}(1+o(1))$ gives $\sum_L \mathbb{P}(\psi\circ T^L < -2\log L) < \infty$, so by Borel–Cantelli $\psi(T^L\alpha) \ge -2\log L$ eventually a.e. The product form follows by exponentiating and absorbing the index shift $L \mapsto L+1$. ☐

Independently of the martingale route, the constant $\pi$ is excluded by the tail calculus that any LIL proof must supply: with variance $\pi^2/4$, exceedance probabilities at threshold $(\pi-\varepsilon)\sqrt{L\log\log L}$ are $(\log L)^{-(\pi-\varepsilon)^2/(2\sigma^2)} = (\log L)^{-2+o(1)}$ at Gaussian scale, and dyadic block maxima then have summable probabilities, so by Borel–Cantelli the lower bound (1.7) of [1] cannot hold with the constant $\pi-\varepsilon$ infinitely often.

## 9.3 The source of the error, located in the authors' own text

Theorem 1.2 of [1] (arXiv numbering: Theorem 2, label `th_theorem2` in the authors' source file, of which the repository now holds a copy at `docs/papers/arXiv-1502.06738v2/Lacunary_products.tex`) is derived in the paper's Section 4 from its Lemma 4.1,

$$\limsup_{L\to\infty} \frac{\sum_{\ell=0}^{L-1} f_1(2^\ell\alpha)}{\sqrt{2L\log\log L}} = \frac{\pi}{\sqrt{2}} \quad \text{a.e.}, \qquad f_1(\alpha) = \log|2\sin\pi\alpha|,$$

which rests on Fortet's classical sharp LIL (the paper's Lemma 2.6): $\limsup S_L/\sqrt{2L\log\log L} = \sigma_f$ with $\sigma_f^2$ given by the functional (2.9) evaluated in theorem 9.1. The paper evaluates the functional in its equation (4.5):

$$\sigma_{f_1}^2 = \sum_{j=1}^{\infty}\Big(\frac{1}{j^2} + 2\sum_{r=1}^{\infty}\frac{1}{2^r j^2}\Big) = \frac{\pi^2}{2}. \tag{4.5}$$

Comparing term by term with the proof of theorem 9.1: the displayed sum is exactly $2c_0 + 2\sum_r 2c_r$, i.e. each Parseval factor $\int\cos^2 = \frac{1}{2}$ has been replaced by 1. The correct value is $\pi^2/4$; hence Lemma 4.1's constant is $\pi/2$, and Theorem 1.2's is $\pi/\sqrt{2}$. Two internal cross-checks against the paper itself: its equation (1.13) states the classical i.i.d. benchmark $\limsup|\sum \mathrm{e}^{2\pi i h X_k}|/\sqrt{2N\log\log N} = 1/\sqrt{2}$ *correctly* (variance $\frac{1}{2}$ of a single harmonic retained), and applying the (4.5)-style evaluation to the single harmonic $f = \cos 2\pi\alpha$ would give $\sigma^2 = 1$ and hence a Hadamard-lacunary LIL constant 1 in the $\sqrt{2L\log\log L}$ normalization, contradicting the classical Erdős–Gál value $1/\sqrt{2}$ which Fortet's formula reproduces. The error is therefore an isolated slip in the evaluation (4.5), identical in the arXiv source and in the published version; it is not an OCR artifact (checked against the publisher's PDF page images and the authors' own LaTeX source).

### 9.4 Numerical corroboration

Sampling the doubling map on exact rational orbits $p \mapsto 2p \bmod q$ (odd moduli $q \approx 2^{61}$, uniform residues, so no floating-point orbit corruption at any length) gives

| | measured | (8), $\sigma^2 = \pi^2/4$ | same with $\sigma^2 = \pi^2/2$ |
|---|---|---|---|
| $\mathrm{Var}(S_{20})$, $4 \cdot 10^5$ samples | 45.66 | 46.06 | 92.12 |
| $\mathrm{Var}(S_{34})$, $4 \cdot 10^5$ samples | 80.41 | 80.60 | 161.20 |
| $\mathrm{Var}(S_{1000})$, $4 \cdot 10^6$ samples | 2459.5 | 2464.1 | 4928.2 |

The covariance $\int_0^1 \psi(x)\psi(2x)\,\mathrm{d}x$ was measured as $0.4112311$ against $\pi^2/24 = 0.4112335$. Tail frequencies at $L = 1000$ track the $\sigma = \pi/2$ Gaussian branch from below (the distribution is left-skewed at finite $L$) and sit far below the $\sigma = \pi/\sqrt{2}$ branch: at threshold $\pi\sqrt{L \log\log L}$ the measured frequency is $5.2 \cdot 10^{-4}$, against $2.7 \cdot 10^{-3}$ for the $\pi/2$ branch and $2.5 \cdot 10^{-2}$ for the branch the printed constant would require. A structural curiosity uncovered along the way: at small $L$ (e.g. $L = 40$) the upper tail is *cut off* by the deterministic Gelfond cap $S_L \leq L \log\sqrt{3} + \mathcal{O}(1)$, which sits below the $c = 2.9$ threshold there; the LIL scale detaches from this cap only for $L \gtrsim 10^2$–$10^3$, which is why naïve small-$L$ simulations cannot see the asymptotic constant at all, and why the corroboration above was run at $L = 1000$ on exact orbits.

*Finding* 9.3 (erratum-level correction to [1]). In Aistleitner–Hofer–Larcher, Ann. Inst. Fourier 67 (2017) 637–687: equation (4.5) should read $\sigma_{f_1}^2 = \pi^2/4$; Lemma 4.1 should read $\pi/2$ in place of $\pi/\sqrt{2}$; the constant $\pi$ in Theorem 1.2 (equations (1.6)–(1.7)) should be $\pi/\sqrt{2} = 2.2214\ldots$; and the constant $\pi/\sqrt{\log 2} = 3.7735\ldots$ in Theorem 1.1 (equations (1.4)–(1.5)) should be $\pi/\sqrt{2\log 2} = 2.6682\ldots$. The paper's $\sigma_{f_2}^2 = 0$ (equation (4.6)), its Theorem 1.3 (whose exponent involves only $\lambda$), and Lemma 5.1's enclosure for $\lambda$ are unaffected. The publisher's PDF (read as page images), the repository's OCR copy, and the authors' own arXiv LaTeX source (`docs/papers/arXiv-1502.06738v2/`, where the theorem is numbered Theorem 2, as Document 3 cites it) all show the same formulas, so this is not a transcription artifact of any copy. Consequently Document 3's Theorem 4.2 constant $\pi$ must be read as $\pi/\sqrt{2}$, and Document 1's fluctuation section is correct as written.

*Postscript: a corrected edition.* Following this audit, the repository owner prepared a corrected critical edition of the paper, stored alongside the OCR transcription as `AIF_2017__67_2_637_0-corrected.tex`. On the four loci of finding 9.3 it agrees with the audit exactly: (1.4)–(1.5) now read $\pi/\sqrt{2\log 2}$, (1.6)–(1.7) read $\pi/\sqrt{2}$, Lemma 4.1 reads $\pi/2$, and (4.5)–(4.6) carry the Parseval factor $\frac{1}{2}$ (yielding $\pi^2/4$ and 0), with the downstream Section 4 displays updated consistently — checked here by search: no occurrence of the erroneous constants survives anywhere in the edition. The edition goes substantially beyond this audit's scope: it repairs the near-resonance step (3.8)–(3.10), whose printed form divides by a ceiling that can vanish for admissible indices, via a bipartite maximum-degree-two argument; enlarges Lemma 2.1's exponential-moment constant from $2\pi^2/3$ to $\pi^2$ and propagates the coarse constants $29 \to 30$, $30 \to 31$, $85 \to 88$; removes Theorem 2.5's inconsistent evenness hypothesis; and corrects a Fourier-tail summation range in Section 4 along with numerous Section 5–6 transcription-level defects. The spot-checks performed for this audit all pass: the internal arithmetic of the new (3.9) ($\|p\|_2^2 \leq \pi^2/12$ on the diagonal, $S_{\ell_1,\ell_2} \leq \pi^2/(2R)$ off it, and $\pi^2/12 + \pi^2/(2(q-1)) \leq \frac{\pi^2}{2}\frac{q}{q-1}$ for $q > 1$), its coherence with the enlarged Lemma 2.1 bound $\exp(\pi^2\lambda^2 Lq/(q-1))$ (the per-block factor $4\lambda^2 W_i \leq 2\pi^2\lambda^2|\Delta_i|\frac{q}{q-1}$ telescopes correctly), and the final Section 4 conversions to $\pi/\sqrt{2\log 2}$. The Section 5–6 repairs were not independently re-derived here and rest on the edition's own validation.

The meta-lesson for this repository's methodology deserves a sentence: the error was surfaced not by checking any single document against the literature — Document 3 *matches* the literature — but by insisting that two independently produced documents agree with each other, and then measuring the disputed quantity. Faithful citation is not correctness.

# 10 Numerical experiments

All experiments evaluate $\log|f|$ by summing $\log|\mathrm{sinc}|$ over levels $h \le \log_2(\pi x) + 30$ with a quartic Taylor tail (absolute error below $10^{-15}$), and integrate over unit lobes by 24-point Gauss–Legendre panels. Scripts: section B.

## 10.1 Shell means and the constant $A_1$

$\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |f|/g_1$, by direct quadrature and, independently, as $\varrho_1^{-n} \int_0^1 \mathcal{L}_1^n(W)$ by collocation ($N = 40$) with Clenshaw–Curtis integration:

| $n$ | 4 | 6 | 10 | 14 |
|---|---|---|---|---|
| direct | 0.0912636508 | 0.0912662060 | 0.0912661248 | 0.0912661241 |
| spectral | 0.0912636508 | 0.0912662060 | 0.0912661248 | 0.0912661241 |

The two computations agree to all ten digits at every $n$ (they share no code path beyond the evaluation of $H$), and the spectral iteration continues to $A_1 = 0.0912661241$ unchanged through $n = 60$. This simultaneously validates the factorization (2), $\varrho_1$, $\kappa_1$, and the spectral-gap convergence claimed by Documents 3 and 4.

## 10.2 Cesàro means: dyadic convergence and the mantissa profile

Running means $\frac{1}{t-t_0} \int_{t_0}^{t} |f|/(A_1 g_1)$, $t_0 = 8$:

| $t = y \cdot 2^n$ | $y = 1$ | $y = 1.25$ | $y = 1.5$ | $y = 1.75$ |
|---|---|---|---|---|
| $n = 5$ | 0.999992 | 0.964484 | 1.103575 | 1.116546 |
| $n = 9$ | 1.000000 | 0.971221 | 1.087225 | 1.100810 |
| $n = 13$ | 1.000000 | 0.971559 | 1.086373 | 1.099966 |
| transfer-operator limit | 1 | 0.971581 | 1.086317 | 1.099910 |

The dyadic column locks to 1; the non-dyadic columns converge to the predicted profile values (agreement $2 \cdot 10^{-5}$ to $6 \cdot 10^{-5}$ at $n = 13$). The full profile (figure 1) has range $[0.9249, 1.1133]$ on a step-0.025 grid. This is the numerical content of "(Q3) yes, (Q2) no".

## 10.3 The logarithmic mean

$\frac{1}{\log(t/t_0)} \int_{t_0}^{t} |f|/g_1 \cdot x^{-1}\,\mathrm{d}x$ at the same mantissas:

| | $y = 1$ | $y = 1.25$ | $y = 1.5$ | $y = 1.75$ |
|---|---|---|---|---|
| $n = 7$ | 0.093861 | 0.092490 | 0.095987 | 0.096493 |
| $n = 13$ | 0.093861 | 0.093287 | 0.094781 | 0.095032 |

All columns approach the common constant $A_1^{\log} = 0.0938610$, at the expected $\mathcal{O}(1/\log t)$ rate; the dyadic column is already exact because complete shells contribute equally. This substantiates theorem 5.1(iv).

### 10.4 Annular maxima and the Lambert formula

Independent maximization (coarse scan over all unit intervals, then bounded refinement), against Document 2's table and its Lambert approximation $\mathcal{A}_k$:

| $k$ | $\log M_k$ (this audit) | $x^*/2^k$ | Document 2 | residual to $\mathcal{A}_k$ |
|---|---|---|---|---|
| 3 | $-11.251778$ | 1.177288 | $-11.251778$ | $-0.823$ |
| 6 | $-25.991031$ | 1.165501 | $-25.991031$ | $-0.728$ |
| 9 | $-46.956338$ | 1.166810 | $-46.956338$ | $-0.723$ |
| 12 | $-74.157742$ | 1.166649 | $-74.157742$ | $-0.783$ |
| 13 | $-84.611550$ | 1.166676 | – | $-0.814$ |
| 14 | $-95.727351$ | 1.083338 | – | $-0.818$ |
| 15 | $-107.493236$ | 1.083331 | – | $-0.784$ |

Rows 3–12 reproduce Document 2's table to every printed digit. At $k = 14$ the maximizer leaves the mantissa $7/6$ (one doubling to the Gelfond cycle) for $13/12$ (two doublings), exactly the pre-period growth $r = \log_2 k - \log_2 \log k + \mathcal{O}(1)$ predicted by Documents 1 and 2; the Lambert residual stays in a bounded band, consistent with the claimed $\mathcal{O}((\log\log k)^2)$ remainder.

### 10.5 Two-adic valleys

Peak of $|f|$ on $(2^k, 2^k+1)$ against $\frac{H(1)^2}{\mathrm{e}(k+1)} 2^{-k(k+1)}$ (Documents 2 and 4; $H(1) = 0.5537712759$): log-differences $+0.137$ ($k = 8$), $+0.122$ ($k = 12$), $+0.098$ ($k = 16$), decaying roughly like $1/k$; peak locations $0.8804, 0.9180, 0.9378$ against the predicted $1 - 1/(k+1) = 0.8889, 0.9231, 0.9412$. For $m = 3$: differences $+0.156$ ($k = 6$) down to $+0.124$ ($k = 12$), with $|H(3)| = 0.0462022991$. The valley laws are supported as stated (asymptotic, with slow $1/k$-type approach).

### 10.6 Variance and tails

See the table in section 9. Method: exact doubling orbits on odd moduli $q \in [2^{61}, 2^{62})$, $p$ uniform on $[1, q)$; each orbit step is one integer shift-and-mod; sin is evaluated in double precision from the exact rational. This eliminates the two artifacts that invalidate naïve double-precision simulation (orbit collapse after $\sim 50$ doublings, and spurious exact zeros from short mantissas), both of which were observed and diagnosed before switching to rational orbits.

### 10.7 The $\varrho_1$ certificate

Document 4's Appendix A script was executed verbatim (SymPy 1.14): for both bracketing rational functions the numerator and denominator Sturm root counts on $[0, 1]$ are zero and the sign checks at $t = 0$ pass, certifying $0.66126807 < \varrho_1 < 0.66134891$ in exact arithmetic, hence $2.74801262 < \kappa_1 < 2.74818899$. This is the strongest rigor statement about $\kappa_1$ in the corpus, and it is machine-checkable.

### 10.8 Exact spot identities

$I_2(n) = 2^{-n}$ verified to $10^{-12}$ ($n = 2, 5, 8, 11$); the slow-ray identity $|f(2^k \frac{4}{3})| = |f(\frac{4}{3})|(3\sqrt{3}/(8\pi))^k 2^{-k(k+1)/2}$ verified to $10^{-14}$ ($k = 6$); Document 4's peak-ray constant $W_{\kappa_\infty}(2/3) = 0.139129774734829$ reproduced to all digits and its ray identity verified to $10^{-13}$ ($n = 9$); the collocation eigenvalues $\varrho_2 = \frac{1}{2}$ (exact to $10^{-15}$), $\varrho_3, \varrho_4$ as printed in Document 4.

# 11 Relation to the repository corpus

## 11.1 The deleted draft

The *Decay rate conjecture* draft (removed from the working tree; recover with `git show e1cdd6260:".../Decay rate conj` claimed

$$g(x) \approx x^{\log(2\pi)/\log 2}(\log x)^{-1/2}\exp\Big\{\frac{(\log x)^2}{2\log 2}\Big\}, \qquad g(x)f(x) \sim \text{constant},$$

after an opening that asserts the opposite sign for the logarithmic power. The audit confirms the four documents' unanimous verdict: the pointwise claim is false on three independent grounds (integer zeros; iterated-logarithm fluctuations on typical rays; two-adic valleys), no power of $\log x$ appears in any of the correct mean normalizations, and the draft's power of $x$ is exactly the RMS exponent $\kappa_2$ — a meaningful constant attached to the wrong norm. The draft's own derivation sketch (termwise summation of the $\log|\sin|$ Fourier series) is precisely the step that erases the dyadic orbit structure; Documents 3 and 4 show what that structure contributes: the difference between $\kappa_2$ and each of $\kappa_1$, $\kappa_0$, $\kappa_\infty$.

## 11.2 Ties to existing material

The Thue–Morse Formula Atlas already contains the sign law $\operatorname{sgn} f|_{(N,N+1)} = \varepsilon_N$, the finite Fourier product, the Gelfond exponent section (the $L^\infty$ row of the present dictionary), and—as the structural twin of the universal factor—the boundary flatness theorem $\log E(t) = -\log^2(1/t)/(2\log 2) + \mathcal{O}(\log(1/t))$ for $E(t) = \prod_j (1 - \mathrm{e}^{-2^j t})$, including the remark that a bounded log-periodic fluctuation persists at the next order. The mantissa profile of figure 1 is the exact analogue of that fluctuation for the decay problem, with the new feature that here the fluctuation is driven by a *singular* measure. On the Lean side, `FourierProduct/PoissonSummation` already carry the product, the scaling law, and rapid decay; `FabiusLogSquaredAsymptotic` and its neighbors carry the small-argument $-\log^2$ layer.

## 11.3 Formalization targets, ordered by feasibility

1. The exact factorization (2) and the $\kappa$-algebra: finite manipulation of already-formalized objects.
2. $I_2(n) = 2^{-n}$: finite Parseval over the Thue–Morse block, adjacent to `ThueMorseBlockAlgebra`.
3. The Gelfond bound via Document 4's two-step inequality (a single polynomial inequality, `nlinarith` territory) or Document 2's subaction identity (pure algebra).
4. The $\varrho_1$ enclosure via the Sturm certificate: rational positivity on an interval; decidable, and the certificate is already machine-readable.
5. The valley asymptotics (elementary boundary-maximum analysis).
6. The variance $\sigma^2 = \pi^2/4$: Parseval for the Clausen series of $\log|2\sin\pi x|$ plus the geometric covariance sum; would also formally certify finding 9.3.
7. Hardest: Birkhoff/ergodic layers and the Perron–Frobenius gap; import as named hypotheses first, as Documents 3 and 4 recommend.

# 12 Verdict

1. **On the question asked.** The bounded form (Q3) is solved, constructively: $g = A_1 g_1$ of (4). The exact-limit form (Q2) is unattainable for any gauge in the natural (shell-shape-stabilizing,

in particular Hardy-field) class; it becomes attainable along dyadic times, or exactly and for all times under the logarithmic mean. This composite answer — not any single formula — is the correct resolution of [6]; within it, the 2018 answer of Negometyanov (a truncated $q$-Pochhammer representation, useful for bounded-range evaluation) contains no asymptotic law, as the question's author already noted at the time. The two readings the four documents left underdeveloped are now treated in full: the lobe-maxima envelope decomposes into the same four regimes and rejoins the $\kappa_1$ gauge through the peak–integral comparison, with $\Theta_{\mathrm{peak}} = 1.9237$ tying the peak and integral readings together (section 6); and the root-mean-square reading is exactly solvable — closed subleading eigenvalue $-\frac{1}{4}$, alternating $\mathcal{O}(1/x)$ convergence, total-energy identity — yet still carries the dyadic phase obstruction (section 7).

2. **On the four documents.** All are technically serious, and their factual overlap survived every numerical test applied here, in several cases to all printed digits. Documents 3 and 4 answer the question; Document 4 is the most rigorous where they overlap (its $\varrho_1$ certificate passes exact-arithmetic execution); Document 2 is the sharpest on the extremal geometry (its numerics reproduce exactly) and the only one to scope the problem honestly; Document 1 is subsumed by Document 2 except for its fluctuation section — which turned out to be the corpus's single most valuable page.

3. **Errors found.** One substantive: the constant $\pi$ in Document 3's Theorem 4.2, inherited verbatim from Theorem 1.2 of the published [1], whose own equation (4.5) drops a Parseval factor $\frac{1}{2}$; the correct constant is $\pi/\sqrt{2}$ (finding 9.3), as Document 1 states. All other deviations found were of scope, convergence-speed candor, or single-ulp arithmetic.

4. **Recommended dispositions.** Absorb into the frontier volume: the $\kappa_q$ dictionary with the certificate-backed $q = 1$ row; the two-part answer to (Q2)/(Q3) with the mantissa profile; the corrected fluctuation constants; the Lambert-form annular envelope and valley laws with explicit convergence-speed caveats. Annotate the OCR copy of [1] at (1.4)–(1.7), Lemma 4.1 and (4.5) with the correction, per the repository's policy on evident errors in stored papers, and consider communicating finding 9.3 to the authors. Retire the four drafts after absorption, per the inbox policy.

## A Constants

| symbol | value | provenance |
|---|---|---|
| $1/(2\log 2)$ | $0.72134752\ldots$ | universal quadratic coefficient |
| $\kappa_0$ | 3.151496129472319 | $\frac{3}{2}+\log_2\pi$ (exact form) |
| $\kappa_1$ | 2.748070014871334(4) | $\frac{1}{2}+\log_2(\pi/\varrho_1)$ |
| $\kappa_2$ | 2.651496129472318 | $\log_2(2\pi)$ (exact form) |
| $\kappa_\infty$ | 2.359014879111741 | $\frac{3}{2}+\log_2(\pi/\sqrt{3})$ (exact form) |
| $\varrho_1$ | 0.661322602060565(2) | collocation; certified in $(0.66126807, 0.66134891)$ |
| $\varrho_2$ | $1/2$ | exact |
| $\varrho_3$ | 0.394896632370234 | collocation |
| $\varrho_4$ | 0.320194101601104 | collocation |
| $A_1$ | 0.0912661241 | shell-mean limit (direct = spectral) |
| $A_1^{\log}$ | 0.0938610 | logarithmic-shell limit |
| $A_2$ | 0.103008918 | RMS shell constant, $A_2^2=0.0106108372$ |
| $\lambda_2(\mathcal{L}_2)$ | $-1/4$ (exact, double) | proposition 7.1; rest of spectrum $<3\cdot 10^{-4}$ |
| $\lambda_2(\mathcal{L}_1)$ | $-0.13435\pm 0.0500\,\mathrm{i}$ | collocation; $\lvert\lambda_2\rvert/\varrho_1=0.217$ |
| $\Theta_{\text{peak}}$ | 1.92372(3) | $\sum_{\text{shell}}E_n/\int_{\text{shell}}\lvert f\rvert$, $k=9$–$14$ |
| $A_1^{\text{peak}}$ | 0.175569 | $\Theta_{\text{peak}}A_1$ (conjectural limit) |
| $\int_{\mathbb{R}}f^2$ | 0.808873535967972 | two independent computations, proposition 7.3 |
| $L^2$ profile range | $[0.894, 1.192]$ | step-0.025 grid, $n=13$ |
| profile range | $[0.9249, 1.1133]$ | step-0.025 grid, $n=13$ |
| $H(1)$ | 0.553771275888881009534 | 30-digit product |
| $\lvert H(3)\rvert$ | 0.0462022991 | direct product |
| $\sigma^2$ (per step) | $\pi^2/4=2.4674011$ | proved (theorem 9.1); measured 2.4595 at $L=10^3$ |
| LIL constant | $\pi/\sqrt{2}=2.2214415$ | corrected; printed value $\pi$ |

## B Verification scripts

The complete scripts (`stage1.py`–`stage6b.py`, in `verification_scripts/`) are short, self-contained NumPy/SymPy programs; `stage6.py`/`stage6b.py` produce the lobe-maxima statistics, the $\mathcal{L}_2$ spectrum, the RMS constants and profile, the Plancherel check, and the $\kappa_q$ table of sections 6 and 7. The essential kernels follow.

Listing 1: Evaluation of $\log\lvert f\rvert$ and the exact-orbit variance measurement

```
def log_abs_f(x):                       # sum over levels h with quartic tail
    H = ceil(log2(pi*max(x))) + 30
    return sum(log|sinc(pi*x/2**h)| for h in 0..H)   # |u|<1e-4: -u^2/6-u^4/180

# variance of S_L on exact rational orbits (no float orbit error):
q = 2*randint(2**60, 2**61) + 1         # odd modulus ~2^61, per sample
p = randint(1, q)                       # uniform residue
for j in range(L):
    S += log|2 sin(pi*p/q)|
    p = (2*p) % q                       # exact doubling
# Var(S_1000) = 2459.5   vs   1000*pi^2/4 - pi^2/3 = 2464.1
```

Listing 2: Transfer operator: Perron root, shell constant, and mantissa profile

```
x[j] = (1 - cos(j*pi/N))/2              # Chebyshev-Lobatto on [0,1]
L1 = 0.5*( diag(sin(pi*x/2)) @ Bary(x -> x/2)
```

```
        + diag(cos(pi*x/2)) @ Bary(x -> (x+1)/2) )
rho1 = leading_eigenvalue(L1)          # 0.661322602060565
A1   = lim_n  rho1**(-n) * clenshaw_curtis( L1**n @ W )   # 0.0912661241
F(z) = lim_n  rho1**(-n) * int_0^z W*P_n / (same at z=1)  # profile mass
P(y) = (1 + F(y-1)) / y                # Cesaro limit at t = 2^k * y
```

Listing 3: Document 4's certificate, as executed (passes)

```
F(z) = 1 + 376189/10^6 z - 69093/10^6 z^2 + 15483/10^6 z^3
R(t) = (u F(u) + v F(v)) / (2 F(2uv)),  u = 2t/(1+t^2), v = (1-t^2)/(1+t^2)
for expr in (R - 0.66126807, 0.66134891 - R):
    assert numerator, denominator have no roots on [0,1]   # Sturm
    assert positive at t = 0
# certified: 0.66126807 < rho_1 < 0.66134891
```

## References


[1] C. Aistleitner, R. Hofer, and G. Larcher, *On evil Kronecker sequences and lacunary trigonometric products*, Ann. Inst. Fourier (Grenoble) **67** (2017), no. 2, 637–687; arXiv:1502.06738. Local copies: `Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/` (published version, OCR, with editorial notes; the same directory holds the corrected critical edition `AIF_2017__67_2_637_0-corrected.tex`) and `Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/` (authors' LaTeX source).

[2] J. Arias de Reyna, *An infinitely differentiable function with compact support: definition and properties*, arXiv:1702.05442; original 1982.

[3] É. Fouvry and C. Mauduit, *Sommes des chiffres et nombres presque premiers*, Math. Ann. **305** (1996), 571–599.

[4] R. Fortet, as cited in [1], Lemma 2.6 (sharp LIL for $\sum f(2^\ell \alpha)$ with Hölder $f$).

[5] A. O. Gel'fond, *Sur les nombres qui ont des propriétés additives et multiplicatives données*, Acta Arith. **13** (1967/68), 259–265.

[6] V. Reshetnikov, *Asymptotic decay rate of an infinite product of sinc functions*, Mathematics Stack Exchange question 2745497 (April 2018); OCR copy in `drafts/rvachev_up_fourier_decay/`.

[7] V. Reshetnikov, *Extrema of an infinite product of sinc functions*, Mathematics Stack Exchange question 2742261 (2018).